SUPPORTING 3D PRINTING

Final Report – Sinclair 2 (LightBeam)

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2 Acknowledgements

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3 Executive Summary

For our capstone project, we worked with a Vancouver-based ultra-fast 3D printing company that designs stereolithography-digital light processing (SLA-DLP) printers. In SLA-DLP printing, ultraviolet light is used to cure liquid photopolymer resin, layer by layer, to create a 3D part. The sponsor company is a leader in SLA-DLP printing, with patented technology that allegedly allows them to print faster than their competitors. During the printing process, supports structures must be added to the part to prevent structural failure. The current process for adding support structures is based on empirical knowledge as it does not account for changing materials or part geometry, and parts are likely over-supported as a result. The goal of our project is to provide recommendations to our sponsor company on how to quantitatively optimize their current support structures by eliminating unnecessary supports while maintaining the sponsor's standard for successful prints. This report captures our progress over the past few months and describes our recommendations for future work.

Our team has limited the scope of our project to developing scientific rules for when supports need to be placed. We have focused on the case of a simple cantilever beam and specified what dimensions result in two main failure criteria: failure by deflection caused by self-loading and failure by separation forces. For both cases, calculations have been completed and, based on these calculations, test pieces have been created. Deliverables to the sponsor company include recommendations, based on our progress thus far, and suggestions for future work if they plan to carry on this research.

Our main recommendation is that in order to reduce support material, improvements to the support generation software are necessary. The current practice of selecting parameters such as maximum overhang length and critical angle is leading to parts being over-supported. Software should be updated to include the effect of part geometry and material properties in the support placement selection process.

4 Problem Definition

Our sponsor company produces printers that employ inverted stereolithography digital light processing (SLA-DLP) printing, which is a form of additive manufacturing [1]. SLA-DLP printing involves projecting a 2D image into a vat of liquid photopolymer resin with ultraviolet (UV) light. The UV light imparts a specific energy into the liquid resin, partially curing the projected 2D image. To build the 3D object, the build platform moves upward the height of one layer to allow room for the next 2D layer. Figure 1 describes the various components of a typical SLA-DLP printer. The object is only partially cured by the printer to allow each layer to fully bond to the previous layer. When the part is in this partially cured state, it is known as the "green" state [2]. Subsequent to printing, parts must undergo a post-cure process to obtain their full strength.

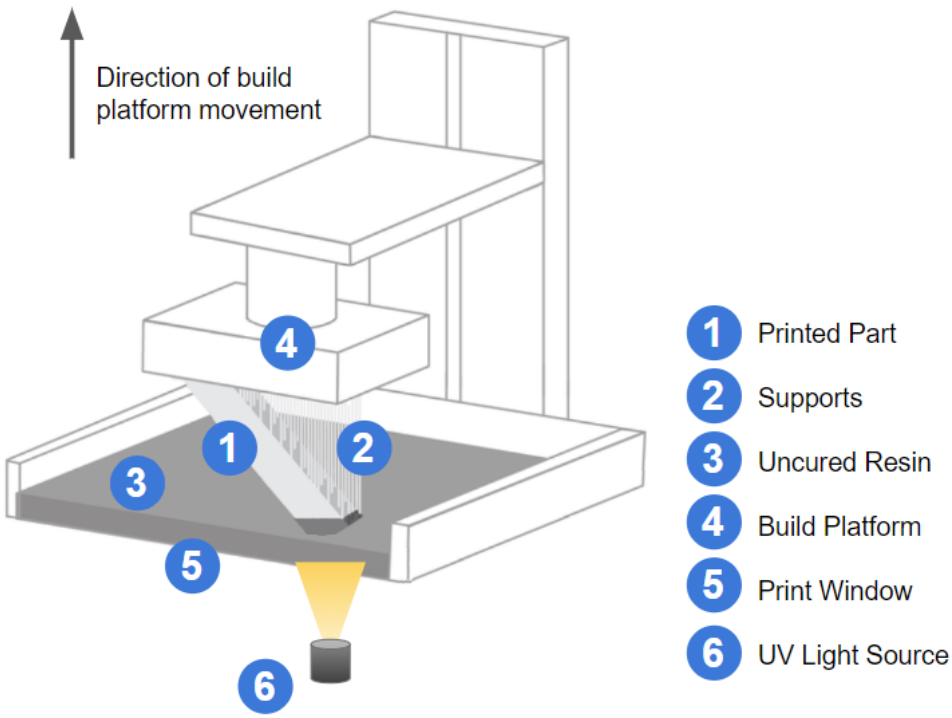

Figure 1: Diagram of a typical inverted SLA-DLP printer [3].

Relatively high-resolution parts can be created using this process – 20 µm compared to 200 µm for typical consumer 3D printers [4] – an advantage that is moving SLA-DLP towards a more established manufacturing process and away from its roots in the rapid prototyping field [5]. However, since it was invented in the early 1980s [6] and has had little time to grow an established field of research, SLA-DLP

printing companies like our sponsor company still rely heavily on empirical knowledge and trade secrets to drive their success [5].

Our sponsor company is on the verge of producing their first SLA-DLP printers for commercial use. They claim their patented technology, a type of membrane that separates the print window from the uncured resin, allows them to print photopolymer parts at record speeds [1]. For example, a part that would take 690 minutes to print using a conventional SLA printer would only take 4.5 minutes to print using the sponsor company's printer [7]. Their technology achieves this quicker print time by reducing the separation force experienced by the part due to the nature of the printing process and thereby eliminating the need for mechanical separation steps usually required for conventional SLA printing. For a conventional printer, each 2D layer is cured directly above the print window, causing the printed layer to adhere to the window [8, 9]. When the build platform is lifted to make room for the next 2D printed layer, the part is pulled off the print window which imparts a downward pulling force on it. This force will be discussed in more detail in *Section 5*. To print the next 2D layer, a fresh layer of resin must exist between the part and the print window and is typically achieved mechanically by sweeping fresh resin into that space. Our sponsor company reduces the magnitude of the separation force and eliminates this mechanical step by introducing an intermediate wettable membrane [10]. The membrane creates a "dead zone" of resin that exists between the printed layer and the membrane/print window.

By using this membrane, the sponsor company has an edge in industries where being able to rapidly prototype objects is essential. One such application would be in hospitals, where doctors could rapidly create organ prototypes for patients. To make the next leap in 3D printing, empirical knowledge will need to be replaced by a quantitative understanding of design, processing, and the product.

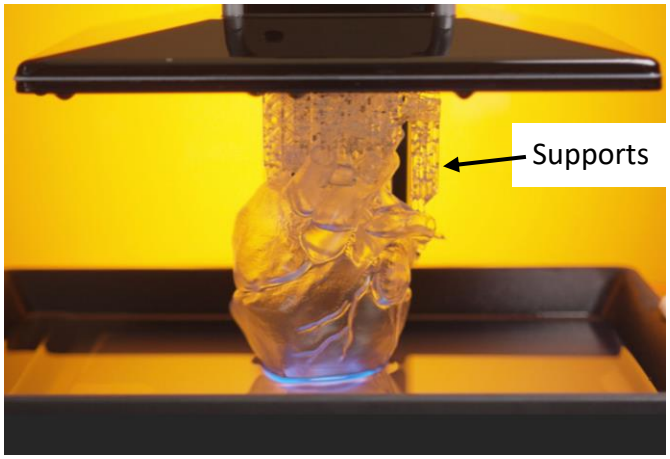

Figure 2: Example of a SLA-DLP-printed organ with support structures [11].

One area that will need quantitative validation is the structural supports for the printed parts. To achieve high-resolution, complex shapes, supports must be added to the design to ensure the geometrical integrity of the parts. An example of such supports is shown in Figure 2. While various methods exist, supports are generally added automatically to the design using software that slices the 3D part into 2D layers. If more supports are added than necessary, this can waste both time and money [12]. The environmental and economic benefit of reducing supports is discussed in *Section 9*. The supports cannot be melted and re-used as the photopolymer material undergoes an irreversible curing process [13]. Further, removing supports during post-processing can result in defects where the supports were connected to the surface of the print. Therefore, the more supports that need to be removed during post-processing, the greater the risk of defects on the final product.

The most general version of our project goal is to create guidelines for the sponsor company in optimizing support structures for their printing [5]. To define the boundary conditions for the design of our project, we have created objectives, constraints, and free variables, which will be discussed in *Section 6*.

5 Technical Review

5.1 SLA Printing

Before printing even begins, a simple way to prepare a 3D part for printing is to determine its orientation – this is a precondition that will determine how many supports are required [14]. While there are several programs that can select the part orientation that results in the least amount of required supports (depending on the type of printer, the part's size, the print material, etc.), this is outside the scope of our project. Regardless of the printing orientation, supports will almost always be required; it is this placement of supports which will be the focus of our report.

The next step in SLA printing is to look at support placement during printing, which will be affected by various failure criteria. Based on discussions with the lead materials engineer from our sponsor company, there are two main failure criteria associated with their SLA printing process: failure by deflection and failure by brittle fracture [2]. Technical reviews for these failure criteria are discussed in *Section 5.2.1* and *Section 5.2.2*, respectively.

Currently, to address these modes of failure, software is used to automatically distribute supports. These solutions work well to expedite the printing process; however, they lack quantitative validation, meaning the addition of supports is based on user experience and trial and error. Examples of these programs include Slic3r and Cura, which are both free to use. The sponsor company uses a software that has been customized for their use and is packaged with the printers they sell [15]. Their software accepts several parameters that can be adjusted, such as critical angle, maximum overhang, and separation force [16, 17], which will be defined in the following sections. These parameters control the point at which supports begin to be added to a model and are currently set based on empirical tests and the experience of the 3D printer operator. The sponsor company uses 2 mm as the typical spacing between supports, which is also the maximum allowable overhang, and 30° as the critical angle [16]. Since these parameters are not uniquely determined for the material or the part, they have the potential to be optimized to decrease the number of support structures.

5.2 Case 1: Overhang

To understand how failure criteria affect support placement, we can either focus on the geometry of the printed part or the geometry of the supports themselves. Our team decided that between the two options, focusing on the geometry of the printed part had more potential to reduce support mass

because it decides when a support needs to exist in the first place. Vanek, Galicia, and Benes described part geometries that may require supports by introducing two types of overhangs: point and face [18], which are illustrated in Figure 3 (a). A point overhang is a point that is lower than its neighbouring points, while a face overhang is a horizontal plane.

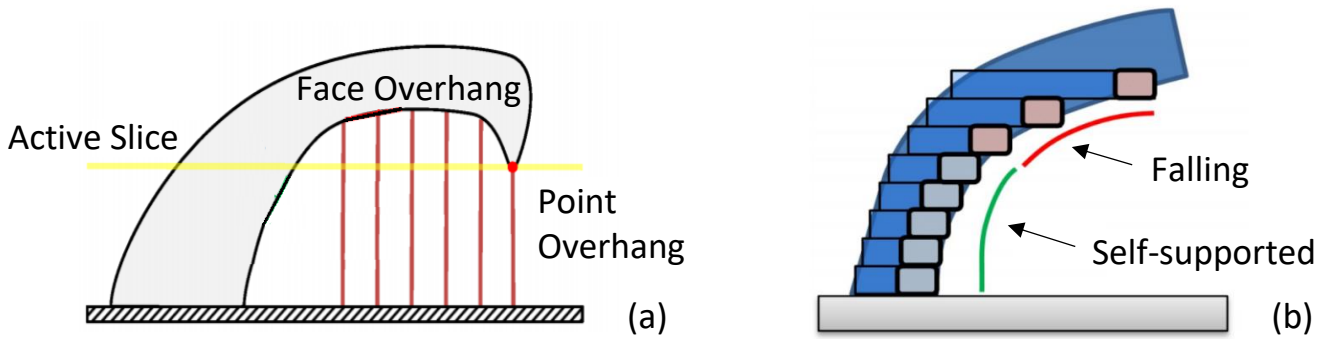

Figure 3: (a) Types of overhangs proposed by Vanek, Galicia, and Benes [18]. (b) Curve geometry as a series of cantilever beams [19].

Simplifying the geometry even further, Livesu, Ellero, Martinez, et al. noted that any SLA-printed surface is simply a stack of cantilever beams, as shown in Figure 3 (b). A cantilever beam is the simplest building block for any geometry that extends beyond the printed layer beneath it. Based on this, we will examine the two failure criteria using the case of a simple cantilever beam.

5.2.1 Deflection from Self-Weight

The first failure criterion we consider is the deflection of a cantilever beam due to self-weight. While the part is submerged in the resin, self-weight is not a great concern since the natural buoyancy of the cured part cancels part of the self-weight, as shown in Figure 4, making it nearly neutral [20]. However, once the part is taken out of the resin, i.e., after the green part has finished printing, self-weight plays more of a role. The sponsor company does not have explicit criteria for failure by deflection, and standards for failure will vary depending on the industry that uses the printer. Due to this, we need to create our own definition of failure, which will be further defined in *Section 6*.

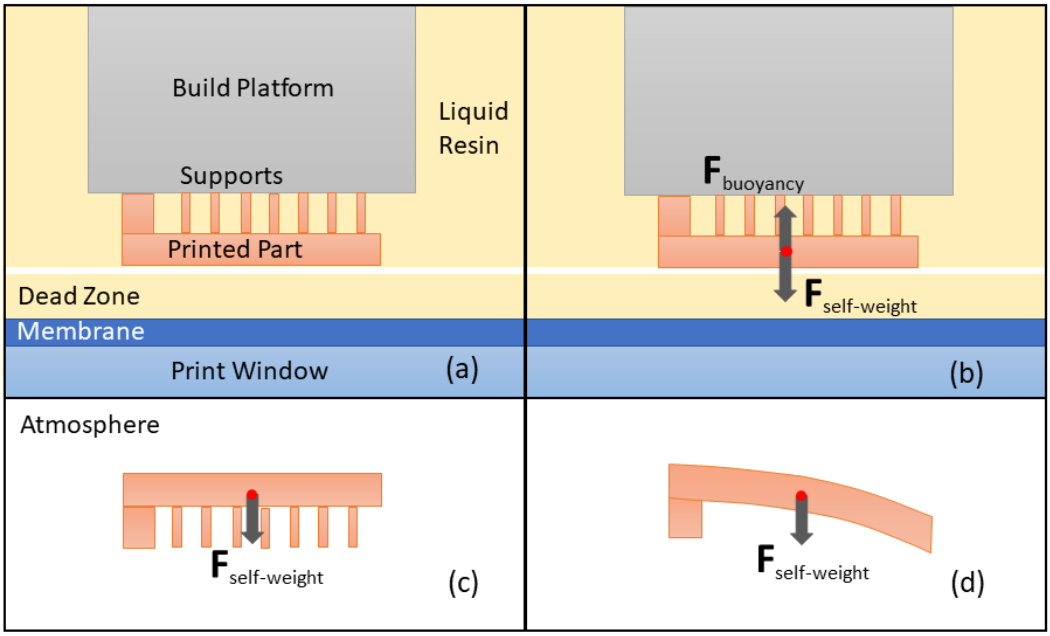

Figure 4: Deflection of a cantilever beam: (a) A cantilever beam is printed. (b) While in the liquid resin, the part experiences negligible self-weight force due to an opposing buoyancy force. (c) If sufficiently supported, when the part is finished printing and is taken out of the liquid resin for a post-cure, it will not deflect. (d) However, if not properly supported, the part may sag due to self-weight.

To calculate the maximum deflection of a cantilever beam under self-weight, we used [21, p. 479]:

$$\delta_{max} = \frac{\mathrm{Fc}^3}{NEI} \tag{1}$$

Where *F* is the weight of the beam, *c* is the length of the beam, *N* is a geometry dependent constant, *E* is the elastic modulus, and *I* is the moment of inertia. We based the material constants, such as modulus of elasticity and density (ρ), on a report jointly produced by the University of British Columbia (UBC) and the sponsor company [1] on a photopolymer resin called P-121. The results from this report are shown in *Appendix A – Mechanical Properties*. Since the printed parts are partially cured to a green state, the green properties of the material must be used for our calculations. To obtain the green state elastic modulus, we can take the elastic modulus at a cure time of zero minutes, which is estimated to be approximately 1.8 GPa. The density of the P-121 resin was provided by the sponsor company [15] and is 1080 kg/m^3.

5.2.2 Brittle Fracture from Separation Forces and Self-Weight

The second criterion for failure is brittle fracture, which can occur if the stress imposed on the part from raising the piece exceeds the brittle fracture stress of the cured resin [22]. Brittle fracture, as opposed to

plastic deformation, is assumed to occur since the yield stress of P-121 is very close to the fracture stress [22]. Also, assuming brittle failure would simplify our calculations. The maximum force experienced by the part is the sum of the force from self-weight and separation force. Self-weight, which has been discussed in the previous section, may be neglected due to the buoyancy force.

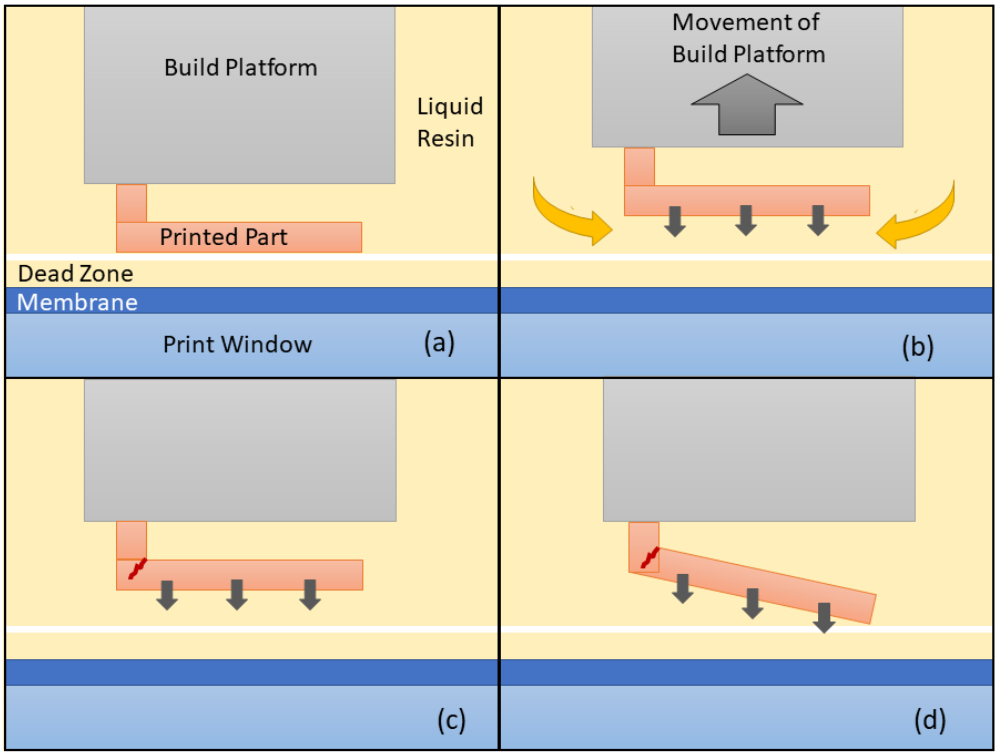

Figure 5: Brittle fracture of a cantilever beam: (a) During SLA-DLP printing, each layer is printed above the dead zone. (b) To make room for the next layer, the build platform moves up. As this happens, liquid resin rushes in from either side to fill the space. This movement imparts suction on the printed layer, known as separation force. (c) If the force is large enough, it will fracture the part at areas of stress concentration, (d) eventually resulting in failure.

The stress in a cantilever beam is calculated using the following equation [21]:

$$\sigma = \frac{F_f c}{N Z_p} \tag{2}$$

Where F_f is the force at failure, c is the length of the beam, N is a constant, and Z_p is the moment of inertia. From this equation, we can see that the stress is dependent on the geometry of the beam as well as the force at failure.

To determine the point at which the part would fracture, we need the fracture stress of the material. The sponsor company provided values for fracture stress in a report on the mechanical properties of P-

121 resin [1]; their results are provided in *Appendix A – Mechanical Properties*. When the printed part is in its green state, the fracture stress is approximately 31 MPa.

We also require the force at failure. The sponsor company compared the separation forces on their patented membrane against a traditional SLA-DLP print membrane [23]. They assumed that separation forces are governed by Stefan adhesion (3), which is adhesion between flat disks immersed in Newtonian fluids [24]:

$$F = \frac{3\pi\eta v r^4}{2h^3} \tag{3}$$

Where η is viscosity, v is separation velocity, r is the radius of the disk, and h is the distance between the disks.

To assess whether we could use Stefan adhesion to predict separation force, we used the same constants as the sponsor company in their report, used equation (3) to compute the force, and compared our results. We found that the Stefan adhesion forces did not match the empirically measured forces. Graphs for our team's calculated values and the sponsor company's empirical values are listed in *Appendix B – Stefan Adhesion*. The most likely reason for this mismatch is the fact that Stefan adhesion is meant for Newtonian fluids, and the uncured resin is a non-Newtonian fluid [24]. Other factors might include the assumption by Stefan adhesion that the force occurs at a right angle to the surface plane, which is not true in practice.

Since this theory did not match empirical testing, we began looking into alternative adhesion theories, but did not find a theory that matched our scope of work. Some methods required knowledge of theoretical values that were not easily obtainable, such as surface energies [25]. Others assumed that the separation was between two solid planes [26, 27], which would have been ill-suited for the sponsor company's setup of a permeable membrane and a partially cured resin part. Ye, Venketeswaran, Das, and Zhou, for example, described the separation using Cohesive Zone Model, which predicts the initiation and propagation of a crack between two rigid surfaces. We decided that continuing down the path of researching separation forces would devolve into a study attempting to prove one theory or another rather than achieving our goal of guiding the sponsor company in the placement of supports. Therefore, we decided that the empirical separation force data previously collected by the sponsor company would be an acceptable estimation but could be a potential source of error if our results do not match our calculations.

5.3 Case 2: Critical Angle

The next step after looking at the simple cantilever beam case is to model beams that are printed at an angle to the build platform to see when supports are required. The angle at which a support needs to be added to a part is known as the critical angle, as can be seen in the figure below. When investigating the critical angle case, there is little current modelling of where supports should be placed [20]. In general, guidelines or rules of thumb that rely on empirical testing are used for determining when to add a support. Currently, the sponsor company prints supports when a part is angled at 30°, regardless of part geometry or material [16].

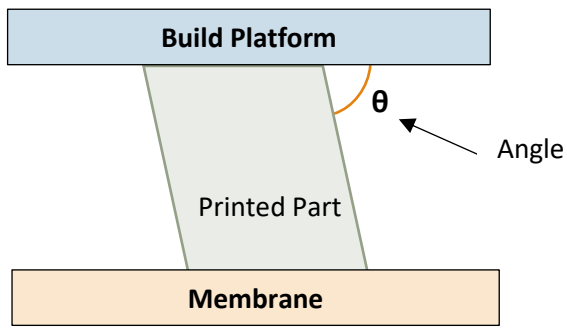

Figure 6: Diagram of critical angle in reference to the build platform and membrane.

In the critical angle case, the maximum force experienced by the part is the sum of the force from self-weight and separation force. Self-weight is considered since the part is hanging off the build platform and could be outside of the resin vat while the other end of the part is still printing. The separation force is induced when the bottom layer of the part is removed from the build platform during the printing process. In reality, the separation force is dominant, since the self-weight adds a negligible amount to the induced stress. Since the separation force is what causes failure, the critical angle case fails by brittle failure. The goal now is to develop an equation or model that can relate the stress in the beam to the geometry of the part and print angle. Further sections will discuss the design options and calculations done to create this model.

6 Problem Objectives and Quantitative Goals

6.1 Objective, Constraints, and Free Variables

Our objective is to minimize the number of supports, thereby minimizing and cost and environmental impact, while maintaining the sponsor company's standard for print success. As previously mentioned in the Technical Review, we decided to approach the challenge of minimizing supports by determining when supports are needed on the printed part. Since the sponsor company does not have explicit criteria for print failure, we have defined our success in a couple different ways, depending on the failure mode. For deflection, we use the amount of deflection that is allowed by current support placement. Since equation (1) is inversely proportional to the thickness of the beam, the largest deflection occurs at the smallest beam thickness. Therefore, the maximum deflection occurs at a layer thickness of 25 µm, with support placed 2 mm apart [2]. Failure by brittle fracture was defined by comparing the maximum stress the part experienced against the fracture stress of the part in its green state. If the maximum stress exceeded this fracture stress, the part was considered to have failed.

Constraints, which specify limitations on the objective, include hard constraints, which are non-negotiable conditions that must be met, and soft constraints, which are negotiable conditions that would be desirable to meet. Our constraints are included in Table 1, with hard constraint values provided by the sponsor company [15].

Table 1: Hard and soft constraints for the objective of minimizing supports.

Hard Constraints	Soft Constraints
• Size of print bed: 19 cm x 12 cm • Material: P-121 • Resolution of the printer: 25 µm vertical resolution, 75 µm horizontal resolution • Dimensional tolerance of part ○ Maximum deflection at 2 mm overhang • Temperature: Ambient (21°C) • Geometries of test pieces: Failure criteria specified as discussed	• Number of supports • Print orientation • Interface (cross-sectional area) between supports and part • Support geometry (shape, thickness, cross-sectional area, fill density) • Separation velocity • Intensity of UV light

Free variables include parameters that can be adjusted to meet our design requirements. To limit the scope of the project, we have selected only one: distance between supports. This was selected as this is the main parameter that can be adjusted in most support placement software and therefore, results based on this parameter can easily be integrated into a design.

6.2 Verification and Goals

To verify if our calculations based on the P-121 resin are correct, test pieces have been created for each failure criteria. The first test piece provides information on failure due to deflection from self-loading. Once printed, we will measure the deflection of the test piece and evaluate its agreement with our calculations. The second test piece provides information on brittle failure due to stress caused by separation force. The two test pieces will be used in tandem to determine the bounds of each failure criterion.

Calculations were also done for the critical angle condition. We built off the work that had been done for overhang to model critical angle. The aim was to identify how the angle at which a specific geometry part is printed affects the amount of supports required.

The economic and environmental impact of reducing support mass was quantified with a socio-economic analysis, discussed in *Section 9.* The analysis provides information to the sponsor company on whether instituting our recommended changes is worth their time and effort. If the environmental or economic benefits are judged to be significant by the sponsor company, this could increase the marketability of their product.

6.3 Assumptions

Throughout our project, we have made several assumptions. We assumed that the only forces acting on the part are gravity, buoyancy, and separation forces between the part and the membrane. Further, the separation forces act perpendicular to the printed layer and act uniformly across the overhangs. Finally, we have assumed that the effects of these forces can be modeled as bending of ideal cantilever beams.

We assumed that all overhangs act as ideal, isotropic cantilever beams, with self-weight distributed uniformly across the overhang. This means that each printed layer is uniformly cured to green strength. In reality, this is not strictly true because the printing induces some anisotropy to the part [22].

7 Design Options

7.1 Failure Criterion

The two failure criteria, deflection and brittle failure, were addressed independently when developing equations and verifying our calculations. The goal of these calculations was to be able to input material properties, such as modulus of elasticity E and density ρ, to determine the maximum printable dimensions that do not require support structures. Based on these calculations, process maps were created to show pass and fail regions of geometry.

Any individual layer which extends past the previous layer can be considered as a cantilever beam. For our calculations, we begun by assuming that the equations for a simple cantilever beam are therefore valid. We used P-121 green resin [28], with E and ρ taken as constants.

7.1.1 Failure Criterion 1: Deflection

Three variables that we can vary for a deflection test piece are the length, width, and thickness of the beam. In this section, we will show that the width of the beam has no impact on the deflection.

The following assumptions are made:

- The cross section of the beam is rectangular;
- Deflection within the elastic region (use the elastic modulus of material);
- The beam is supported at one end and free to deflect at the other end;
- The beam deflects under self-loading only; and,
- The cross section of the beam doesn't deflect out of its vertical plane.

Table 2: Variables and symbols for the deflection of a cantilever beam.

Variable	Symbol	Units
Thickness of beam	a	m
Width of beam	b	m
Length of beam	c	m
Elastic Modulus	E	Pa
Density	ρ	kg/m^3
Gravitational Constant	g	$9.81\ m/s^2$
Maximum Deflection	δ_{max}	m
Load Type Constant	N	-

We began with the equation for maximum deflection under self-weight [21, p. 479], as shown by equation (1). The variables are listed in Table 2 and depicted in Figure 7. Equations for the moment of inertia [21, p. 477] and force are provided in equations (4) and (5), respectively. In equation (5), buoyancy was neglected since we are looking at the deflection of the part after it has been taken out of the liquid resin bath, but before it undergoes its post-cure procedure.

$$\delta_{max} = \frac{\mathrm{Fc}^3}{NEI} \tag{1}$$

$$I = \frac{ba^3}{12} \tag{4}$$

$$F = \rho abcg \tag{5}$$

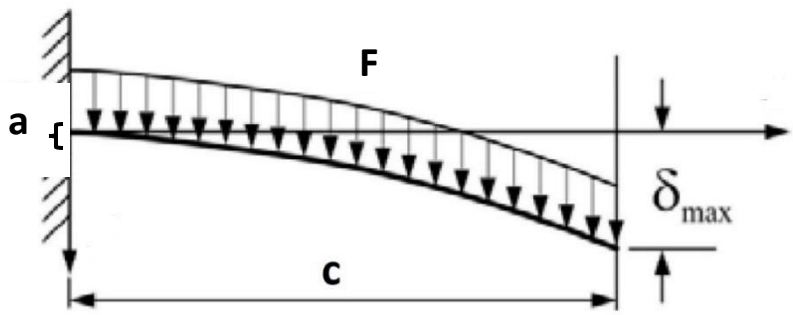

Figure 7: Deflection of a cantilever beam from self-loading [29]

By inserting (4) and (5) into (1), setting the Load Type Constant as 8 as defined by the geometry and simplifying, we arrive at the final equation for deflection of a cantilever beam:

$$\delta_{max} = \frac{3\rho g c^4}{2Ea^2} \tag{6}$$

After calculating the deflection of the beam, we decided to examine case 1: how much deflection is predicted for a certain length *c* overhang beam while keeping the thickness *a* constant. In the derived equation, length *c* is to the power of four, while thickness *a* is to the power of two. Therefore, the length of the beam has a larger effect on the deflection than the thickness of the beam, and varying length will have a greater impact on failure than varying thickness.

From these calculations, a process map of the critical dimensions for self-loading conditions was created, as shown in Figure 8.

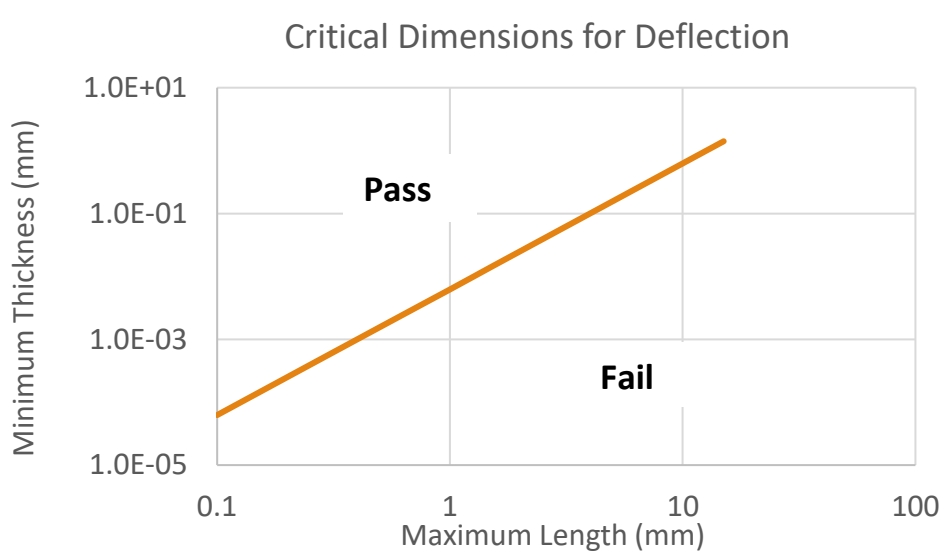

Figure 8: Process map of the critical dimensions for deflection, with the pass region indicated as the area above the line.

Based on this process map, we can see the relationship between critical thickness and length. As the length is increased, the thickness also needs to be increased in an exponential fashion to prevent failure.

7.1.2 Failure Criterion 2: Brittle Fracture

The next step after calculating the maximum length allowable by failure criterion 1, was to compare this to the brittle fracture failure criteria. Both self-loading and separation forces induce stresses in an overhang and must be considered for these stress calculations. As discussed in S*ection 5* of this report, we assumed that the work done by the sponsor company and Professor Sinclair's research will be a close enough analog to the work done in this report to hold the empirical equations they derived as valid.

Table 3: Variables used in stress calculations.

Variable	Symbol	Units
Stress in beam	σ_y	MPa
Sum of forces acting on beam (self-load + separation)	F	N
Load type constant	N	-
Moment of inertia	Z_p	m^4
The difference between the density of the material and the density of the resin	$\Delta\rho$	$\frac{kg}{m^3}$
Forces due to self-weight of the part	F_g	N
Forces due to separation	F_s	N
Diameter of the circular area under separation force	D	m
Width of beam	b	m
Length of beam	c	m
Thickness/height of beam	a	m
Area of the beam	A	m^2
Gravitational constant	g	$\frac{m}{s^2}$

Stress in a cantilever beam may be calculated with the following equation [21]:

$$\sigma_y = \frac{F_f c}{N Z_p} \tag{2}$$

Ideally, this equation should be expressed in terms of the overhang geometry to easily understand the relationship that length, width, and thickness have on stress. To begin, we can resolve the force in the z-direction as the self-weight of the part F_g and the separation force F_s:

$$F = F_g + F_s \tag{7}$$

The gravitational force is the product of the mass and gravitational constant:

$$F_g = \Delta\rho g a b c \tag{8}$$

Separation force can be described with the same approach as used in the previous work done by the sponsor and Professor Sinclair. Using an area of $A = bc$ and simplifying results in:

$$F_s = 0.0707\ exp(90.496\sqrt{bc}) \tag{9}$$

A similar calculation can be found in *Appendix E – Deflection Calculations* for the full derivation of the above equation. Inserting (8) and (9) into (7):

$$F = 0.0707\ exp\left(90.496\sqrt{bc}\right) + \Delta\rho g a b c \tag{10}$$

Inserting (10) into (2), and setting the constant for a cantilever beam supported at one end to its given value, $N = 2$:

$$\sigma_y = \frac{c}{2Z_p}\left(0.0707\ exp\left(90.496\sqrt{bc}\right) + \Delta\rho g a b c\right) \tag{11}$$

Inserting the corresponding moment Z_p [21] p. 477, for a rectangular cross section:

$$Z_p = \frac{ba^2}{4}$$

$$\sigma_y = \frac{2c}{ba^2}\left(0.0707\ exp\left(90.496\sqrt{bc}\right) + \Delta\rho g a b c\right) \tag{12}$$

The entire equation is now in terms of the dimensions of the overhang. Using the brittle failure stress of P-121 and varying the length and width of the cantilever, we can generate a process map using equation

(12). For this process map, the self-loading force term $\Delta\rho gabc$ was neglected from equation (12) because it results in an insignificant stress compared to the stress from the separation forces. Therefore, separation forces dominate the brittle failure mode.

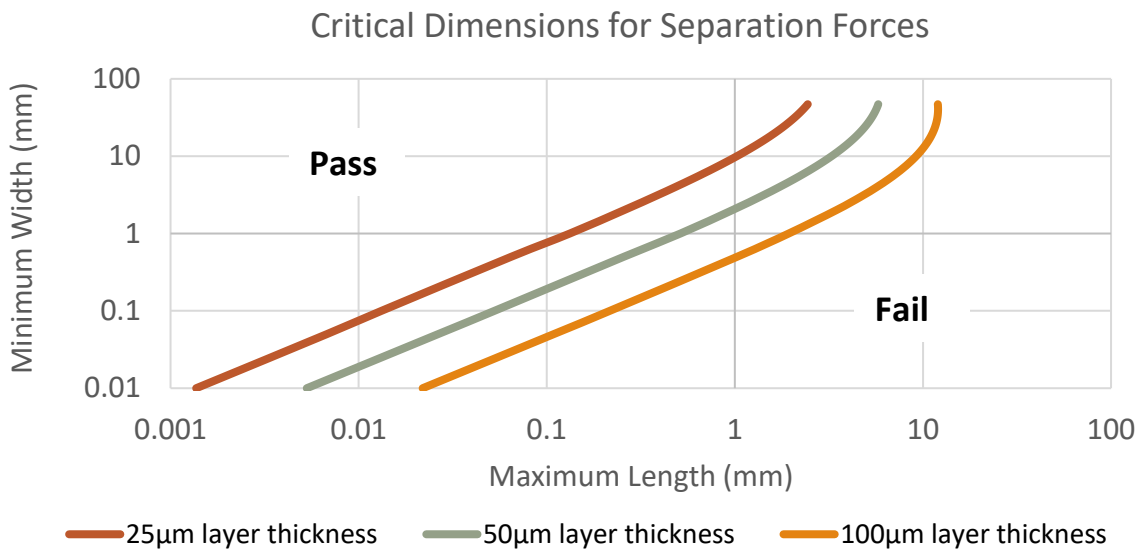

Figure 9: Process map of the critical dimensions for separation forces, with pass regions indicated as the area above each line.

The curves on the process map in Figure 9 show the pass and fail regions for three different thicknesses, which represent the different resolutions at which the sponsor company can print. Any overhang that has dimensions found above the critical stress curves will result in a successful print and any overhang below the critical stress curve will fail by brittle fracture. From the process map, it can be seen that as the length of the part is increased, the width must also be increased. In addition, as the resolution of the printer is decreased, the distance between supports may be increased. These calculations and assumptions were to be tested using designed test pieces, as discussed in *Section 8.2*.

7.2 Critical Angle

After looking at two failure criteria for a simple cantilever beam, the next step was to model beams that are printed at an angle to the build platform to see when supports are required. Currently, the sponsor company prints supports when a part is angled at 30° regardless of part geometry or material [2]. The objective was to see if this guideline is over supporting parts based on various geometries.

We determined two ways to approach creating a model for the critical angle case. Option 1 was to look at the part as a series of cantilever beams. This would require modelling each of the layers as an

individual beam and resolving the forces and moments for each. Then, the individual beams would need to be mathematically combined to solve for the stress on the entire part. We referred to option 1 as a microscopic model because it looks at each layer individually.

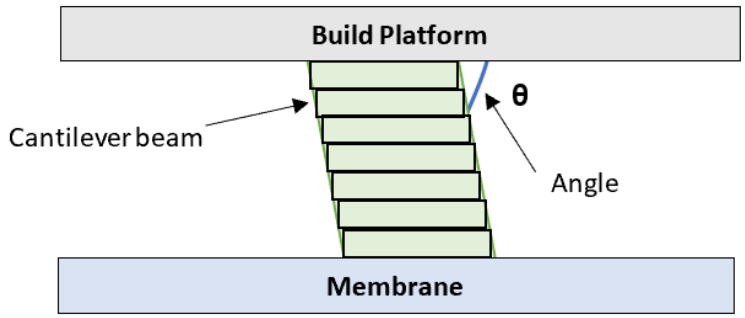

Figure 10: Critical angle microscopic model. Here, a part is modelled as a series of beams.

Option 2 was to look at the part from a macroscopic view and resolve the forces and moments for the entire part. The angle θ at which supports need to be placed is called the critical angle, as can be seen in the diagram below.

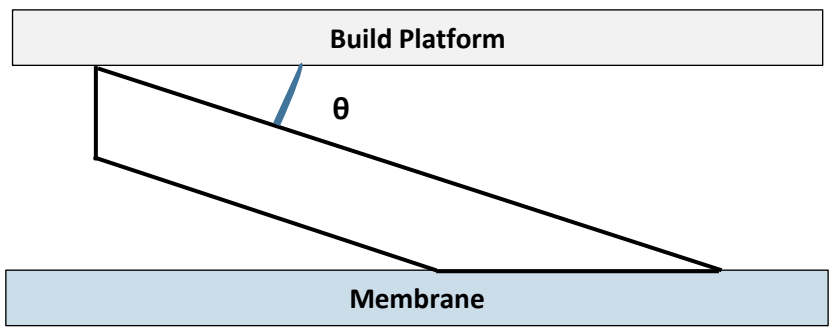

Figure 11: Critical angle macroscopic model

We decided to pursue Option 2, the macroscopic model, since it modelled the part in a way that could be calculated based on our mechanics equations developed before for each failure criteria. The goal was to develop a model/equation that can relate the stress in the beam to the geometry and print angle. This model combines the two failure criteria discussed above to evaluate their effects of the critical angle case. This will be discussed in *Section 8 – Detailed Design and Implementation*.

8 Detailed Design and Implementation

To verify if our calculations and assumptions were correct, we created test pieces for the sponsor company to be printed by their SLA-DLP printer. If the test pieces matched our predicted values, then our calculations and assumptions are valid. If not, we would need to revisit our initial assumptions. The creation of these test pieces is discussed in the sections below.

8.1 Design for Deflection Caused by Self-Load

We have defined failure by deflection as the deflection resulting from a 2 mm overhang (current guideline) at 25 µm thickness (thinnest resolution of the printer). We entered these values into equation (6), our derived equation for deflection of a cantilever beam, to find the maximum allowable deflection. The deflection was calculated to be 0.23 µm, and therefore failure by deflection was defined as greater than 0.23 µm. With failure now defined, the maximum deflection equation was used to determine the geometry of the test piece.

The test piece for deflection needed to meet specific criteria: it needed to either verify or disprove our calculations, fit in the 19 cm x 12 cm print bed, be robust enough to survive transportation from the sponsor company's location to UBC, and provide a range of predicted deflections. To verify our calculations, the test piece should have overhangs with a geometry that results in a measurable, yet moderate amount of deflection. Small deflections would be difficult to measure, and large deflections could result in interference with other parts of the test piece. From equation (6), we can observe that with decreasing overhang thickness, there is increasing deflection. Thus, decreasing the thickness would achieve the first two criteria, since shorter overhangs will fit inside the relatively small print bed. However, the test piece needed to be robust enough to survive transportation and handling to UBC. To satisfy these competing criteria, our team decided that the thickness should be at least 1 mm. Using this thickness and defining lengths of 8 cm, 10 cm, and 15 cm, we predicted deflections of 361.6 µm, 882.9 µm, and 4469.7 µm, respectively. The lengths were chosen specifically to produce deflections about a magnitude apart, verifying our calculations at several points along the predicted pass/fail curve discussed in *Section 7.1.1 – Failure Criterion 1: Deflection*. Schematics of the test piece are shown in Figure 12.

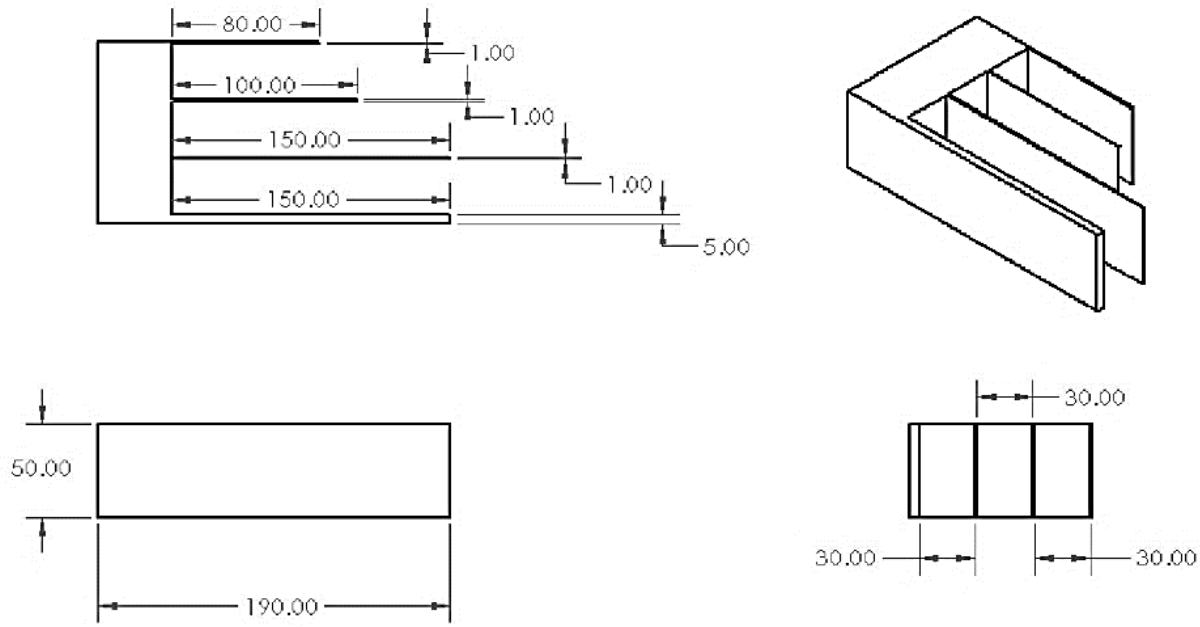

Figure 12: SLA-DLP test figure dimensions, in millimeters.

The test piece is printed with the overhangs vertical in the printer so that there is no interference from supports and minimized risk of damage during printing. This orientation is shown in Figure 13, where the face indicated in blue is attached to the moving build platform.

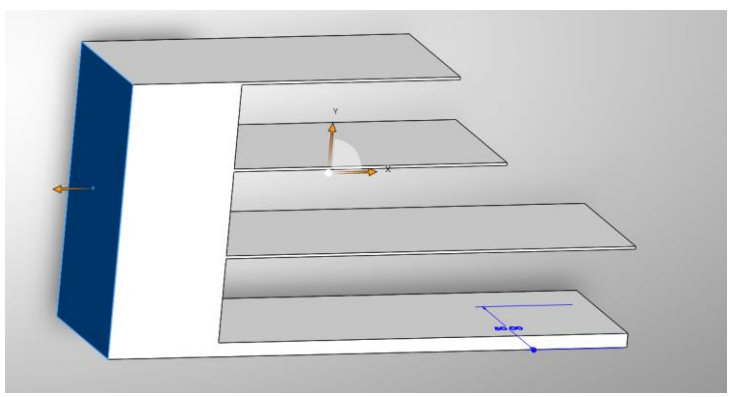

Figure 13: Print orientation of deflection test piece; the blue region is to be attached to the build platform.

One potential concern for this orientation was the large contact area, leading to the potential for large separation stresses. Calculations done in *Appendix E – Deflection Calculations* show that these stresses are less than the fracture stress of the material and therefore there is no risk of fracture from printing in this orientation.

Another factor that was considered is the direction of anisotropy since the x-z plane of testing will now be isotropic while the y-direction is not. According to Professor Sinclair, the elastic properties are very close to being isotropic [22]. Since this test piece is for deflection, which is a function of the Young's modulus, this print orientation is acceptable. In summary, the profile-to-print-bed orientation is ideal due to its lack of supports on critical surfaces and strength to resist fracture from separation force.

Though we developed these test pieces, the sponsor company was unable to print the deflection test piece. As a result, we are unable to verify if our calculations and assumptions are correct. Printing of the test piece is recommended as a future step in *Section 11 – Recommendations*.

8.1.1 Deflection Measurement Procedure

Designing and printing test pieces are one challenge – measuring the deflection that results from the test pieces is another. Being able to measure the deflection accurately and precisely is important to be able to verify or disprove our calculations. Our team decided to create a measurement procedure, using a simplified, fused deposition model-printed (FDM) test piece as a substitute for our deflection test piece. FDM was chosen because it is a rapid, relatively cheap method of creating parts. By creating this FDM analog, we were able to develop a procedure we were confident would adequately measure our final deflection test piece. The following steps outline our procedure:

1. Ensure the following are all level:
    a. Surface on which the test piece stands
    b. Surface on which the camera stands (use tripod)
2. Ensure the area is well-lit (adjust lighting if necessary).
3. Place the test piece in front of paper to provide an even, contrasting background.
4. Position the camera directly in front of the test piece and focus on the test piece.
5. Take the photo.
6. Open image in an image software, such as Image J.
7. Calibrate the scale based on the overhang thickness.
8. Measure the height of the overhang at the base of the overhang.
9. Measure the height of the overhang at the end of the overhang.
10. The difference between these two measurements is the deflection.

We tested this procedure on an FDM-printed, 15 cm-long overhang made from polylactic acid (PLA), and found that we were able to accurately measure our predicted deflection with a smartphone camera. The

small amount of error was assumed to be from the difference between the literature value for Young's modulus of PLA, which is isotropic, and the FDM-printed PLA, which is anisotropic.

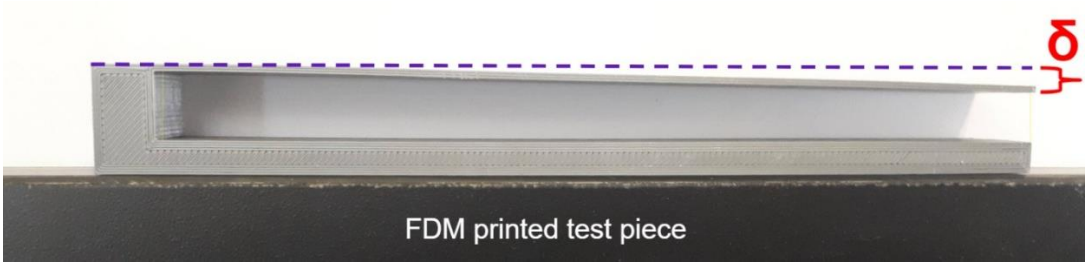

Figure 14: Deflection of the FDM test piece used to confirm our measurement procedure.

Generally, for smartphone images, an algorithm would need to be applied to eliminate geometric distortion, a problem commonly experienced by smartphone cameras [30]. To keep things simple, geometric distortion was minimized by keeping the camera lens level and the test piece in the center of the photo, away from the edges. Using ImageJ, a common image processing software, we were able to measure to a 30 μm resolution, which is well below the minimum predicted deflection by our SLA-DLP test pieces. Therefore, we are reasonably confident that the deflection of our SLA-DLP test pieces will be able to be accurately measured.

8.2 Design for Brittle Failure Caused by Separation Force

The calculations and assumptions discussed in *Section 7.1.2* will be verified using three test pieces. These test pieces, depicted in Figure 15, consist of overhangs that test the pass and fail regions of our brittle failure process map, as shown in Figure 16.

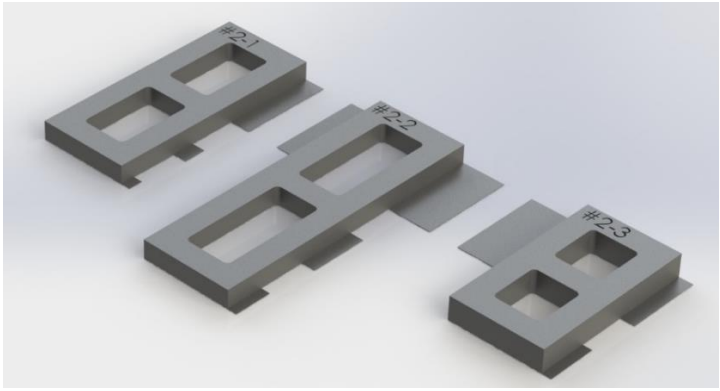

Figure 15: Separation force test pieces.

Like the deflection test piece, the brittle failure test piece needed to satisfy several criteria: it needed to either verify or disprove our calculations, fit in the 19 cm x 12 cm print bed, be robust enough to survive

transportation from the sponsor company's location to UBC, and provide several points of data. Unlike the deflection test piece, where a range of deflections was possible, our calculations predicted either a "pass", i.e., an undeformed overhang, or a "fail", i.e., a broken overhang. Since it was unlikely that our initial calculations would be completely correct, our team decided to create a test matrix of overhangs that would likely pass and likely fail, with some data points chosen between the two regions to assess the accuracy of our model.

First, we considered the dimensions of the overhangs. We chose a layer thickness of 100 µm as this was the thickness used in the original separation forces study discussed in the technical review, and therefore the most likely to behave as expected. It is also the thickness with which the sponsor company usually prints. Next, the length and the widths of the overhangs were chosen. The calculations discussed in *Section 7.1.2* were modified to account for potential stress risers at the corners where the overhangs met the main body of the test piece. Since the stress concentration factor for circular holes is 3 and increases with increasing crack tip sharpness [31], a stress concentration factor the same order of magnitude, but greater, was chosen. A stress concentration factor of 5 was selected, and the resulting change to the brittle failure curve is shown in Figure 16. The red lines show the three sets of overhang dimensions discussed next.

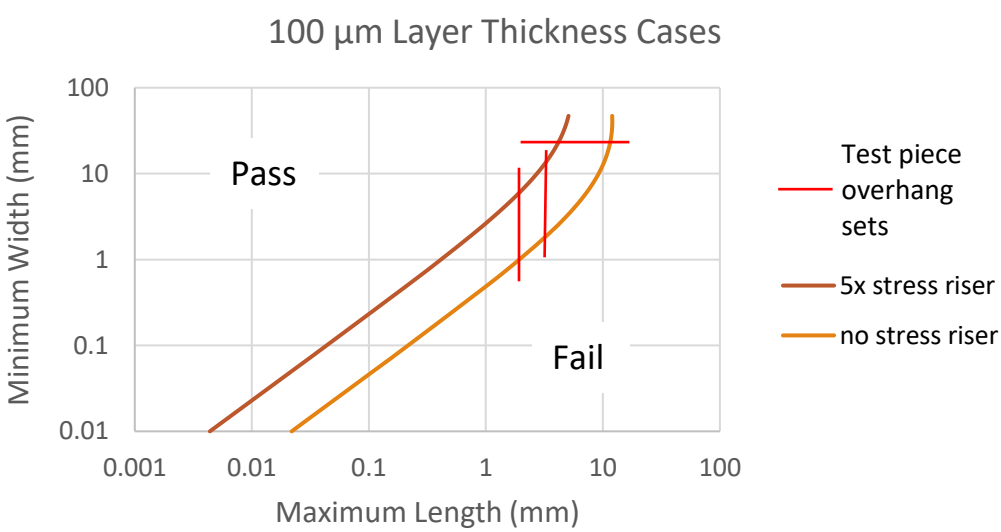

Figure 16: Stress risers in brittle failure model.

We expected that brittle failure would occur somewhere within the region between the two curves as a 5x stress concentration is the largest stress we expect could result from the corner and no stress riser

would result in the lowest stress possible. In order to create a pass-fail matrix, we began with a width of 20 mm, which is in the region where the curves begin to extend vertically, and selected 5 evenly spaced lengths beginning before the 5x stress riser curve and extending to past the no stress riser curve. We expected that some of these points would fail and some would pass. We expected the pass-fail boundary for this case to take place closer to the 5x stress riser, around a 4 mm length.

Next, we selected lengths of 2 mm as this was the smallest we felt comfortable being able to visually identify as passed or failed and a length of 3mm for a third data set spaced between the other two. These lengths were then held constant and 5 widths were evenly spaced from above the 5x stress riser curve to below the no stress riser curve for each case, similar to how the overhang dimensions were selected for the constant width overhangs.

Taking all these factors into consideration, we arrived at 3 sets of overhangs as listed in the table below.

Table 4: Brittle failure overhang dimensions.

Overhang Set	Length (mm)	Width (mm)	Expected outcome
#2-1 Constant Width	2	20	Success
	5	20	Stress riser dependent
	7	20	
	9.5	20	
	12	20	Failure
#2-2 Constant Length	3	1	Failure
	3	4.75	Stress riser dependent
	3	8.5	
	3	12.25	
	3	16	Success
#2-3 Constant Length	2	0.5	Failure
	2	3	Stress riser dependent
	2	5.5	
	2	8	
	2	10.5	Success

From discussions with our sponsor, we decided to add holes through the center of the test piece to reduce the part volume, thereby reducing the material used, and to reduce the separation force on areas not being tested. The final, dimensioned design is shown below.

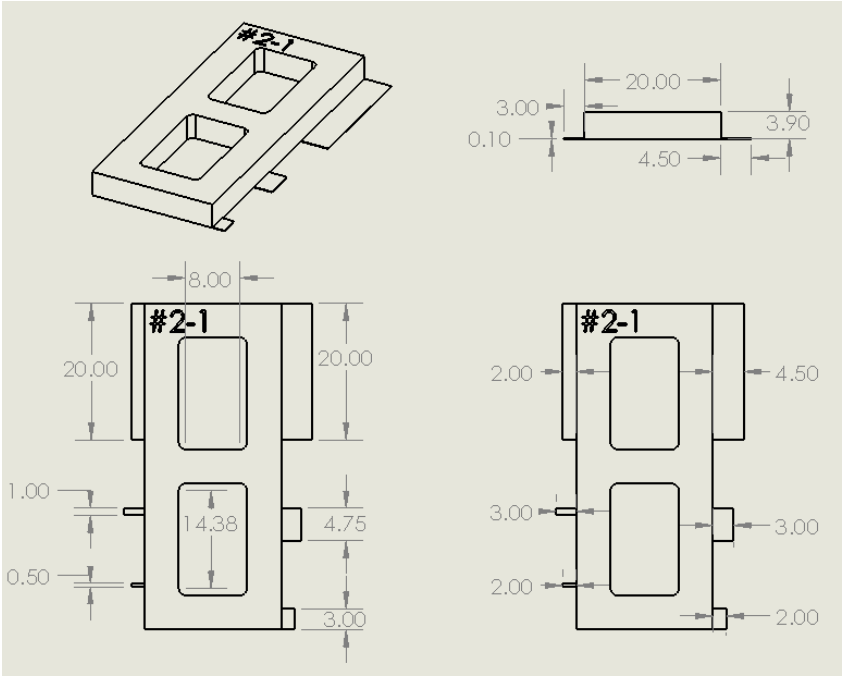

Figure 17: Schematic and dimensions of final separation force test piece #2-1.

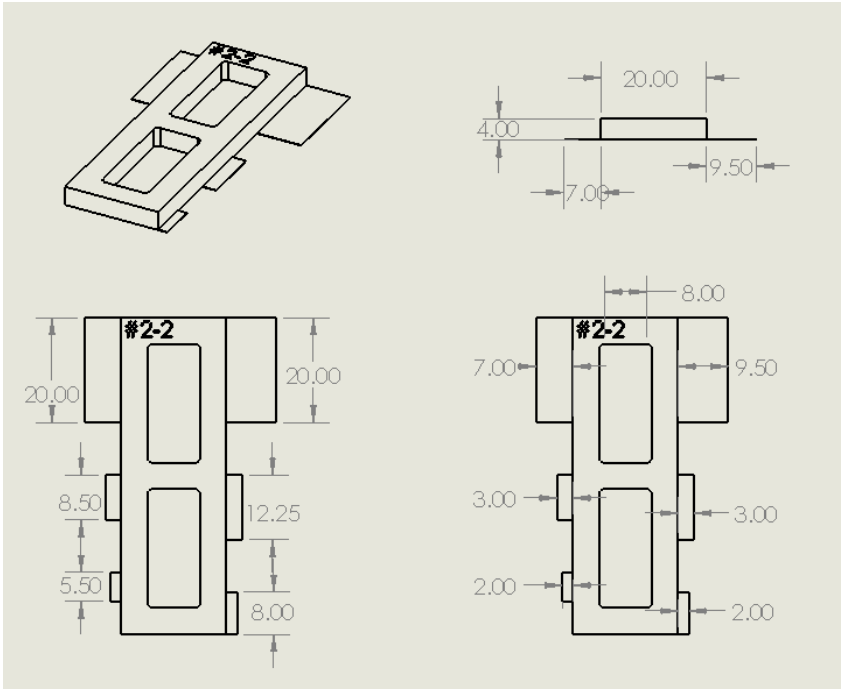

Figure 18: Schematic and dimensions of final separation force test piece #2-2.

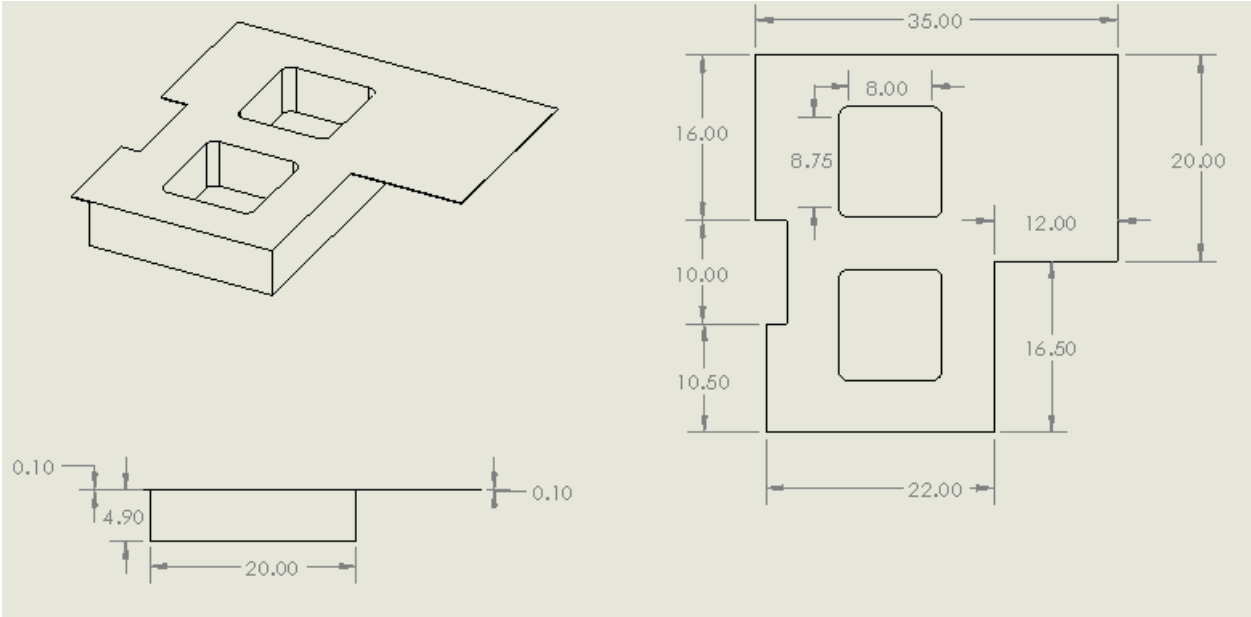

Figure 19: Schematic and dimensions of final separation force test piece #2-3.

Though we developed these test pieces, the sponsor company was unable to print the brittle fracture test pieces. As a result, we are unable to verify if our calculations and assumptions are correct. Printing of the test pieces is recommended as a future step in *Section 11 – Recommendations*.

8.3 Design for Critical Angle

When parts are printed to an angle to the build platform, they require additional supports. The angle at which supports need to be placed is called the critical angle. To model critical angle, the separation force is treated as a point load, while the weight of the beam is constant across the part. The goal is to convert these forces into moments to define the stress in the beam as a function of the support angle.

Figure 20 helps to understand the scenario. The build platform is at the top and the membrane at the bottom with the red arrows indicating the acting forces.

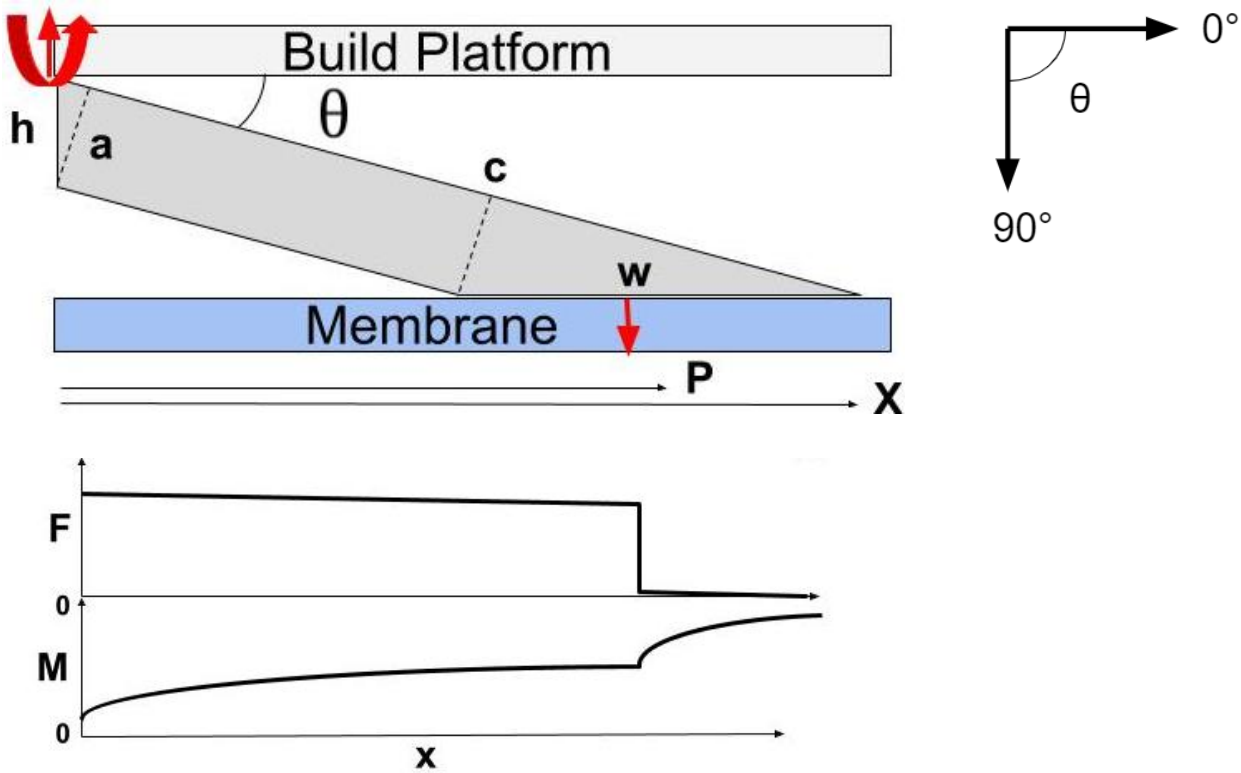

Figure 20: Diagram of critical angle in reference to the build platform and membrane with the calculated forces and moments.

Table 5: Critical angle variables.

Variable	Symbol	Units
Bending stress in the beam	σ_{bend}	Pa
Height of the area on contact with the bulk part	h	m
Width of the area in contact with the membrane	w	m
Support angle	θ	$degrees$
Horizontal distance from the end on the beam to the previous support	X	m
Horizontal distance to the separation force point load	P_s	m
Horizontal distance to the self weight point load	P_g	m
Moment due to the separation force	M_s	$N\ m$
Moment due to the self-weight of the part	M_g	$N\ m$
Section modulus	Z	m^3
Inertia	I	m^4
Distance from centroid	y	m

Since the forces in the beam are no longer acting perpendicular its length, there are two components of stress to be modelled: bending stress and tensile stress. The bending forces act perpendicular to the length of the beam while the tensile forces act parallel. Although the forces act perpendicular to each other, both stresses act longitudinally to the part and so they may be summed.

The top of the beam in Figure 20 is experiencing tension along its longitudinal axis from two stress components, the bending stress and the tensile stress. This is where we would expect to see failure for this mode. It is also likely that the failure would occur at the base of the beam, near position $X = 0$. The calculations for these components can be found in *Appendix F*.

$$\sigma_f = \sigma_{tensile} + \sigma_{bending}$$

The theoretical limitations of this model are due to the geometry. At upper and lower limits of θ the geometry of the angles beam becomes triangular. When the width of the beam in contact with the print membrane w is equal the horizontal distance X, the model is no longer valid. This occurs when $\sin(\theta) = \frac{a}{c}$. Likewise, when the thickness of the beam a is equal to w the model no longer holds true. For a 1 mm thick, 10 cm long beam, the valid range of angles is $\theta = 0.5$ to 89.5 degrees.

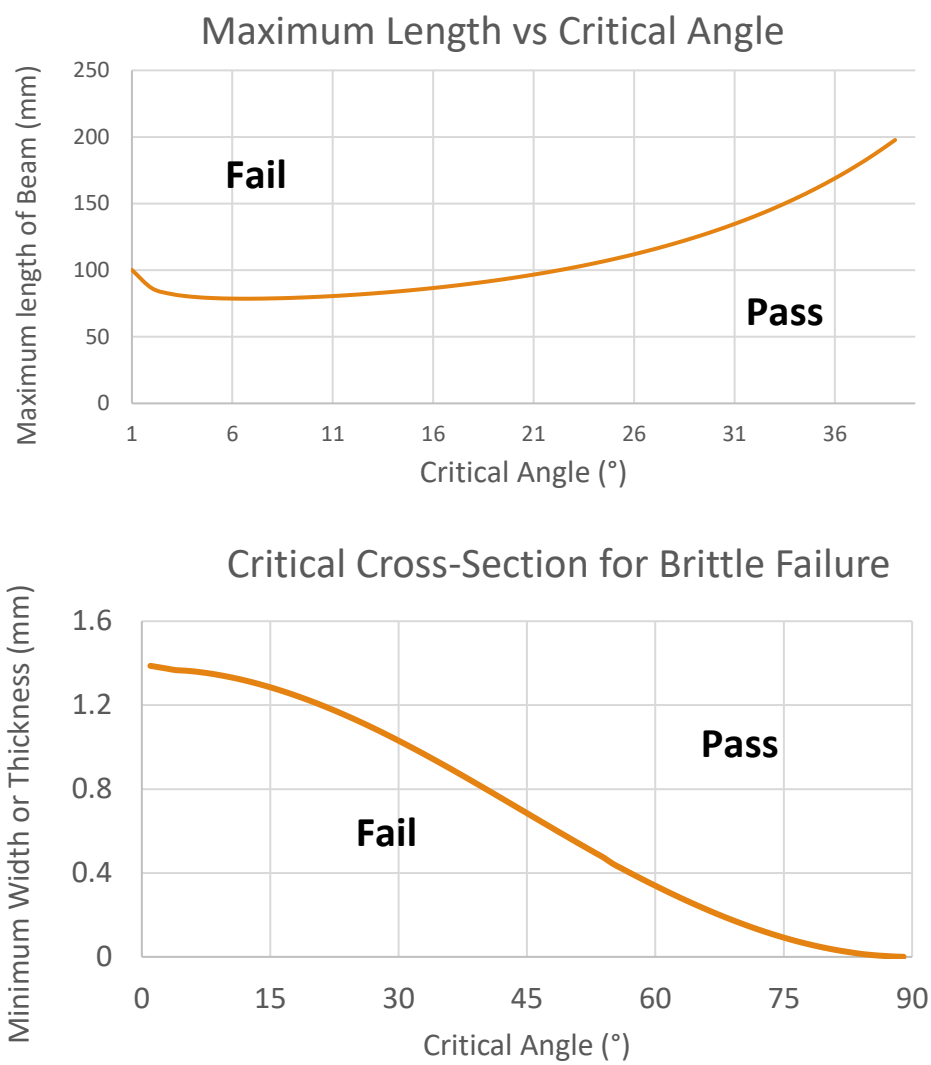

Figure 21: Process maps for the critical angle case.

From the process map, trends are seen between the geometry and the print angle. As we increase the angle, we can print longer beams without support, holding all else constant. In addition, the thickness of the part can decrease as you increase the support angle away from horizontal.

Unfortunately, due to time constraints of the project, our team was unable to develop a critical angle test piece for the sponsor company. Our recommendations for future work on critical angle are summarized in *Section 11*.

9 Socio-Economic Assessment of Design

This socio-economic assessment examines the economic and environmental impact of reducing supports in SLA-DLP printing. The economic impact will quantitatively assess monetary savings realized by the new support design compared to the current design, while the environmental impact will quantitatively compare the CO_2 and energy footprints. A case study was examined to explicitly compare the current practice for support generation to the new model developed from our calculations.

9.1 Economic Analysis

The economic analysis investigated the major factors that impact the cost of SLA-DLP printed parts and how a reduction in support material would affect the overall cost. The cost of a printed part was split into three main categories: material, power, and labour. The economic analysis considered both a small-scale operation and a large-scale operation. The small-scale operation is similar to our sponsor company, with a small number of printers and operators, while the large-scale operation is meant to reflect the process if it were scaled-up to a production line. While production line assembly is not yet common for SLA-DLP printing, an example is the collaboration between Adidas and Carbon 3D in printing shoe soles [32].

The material category includes the cost of the material itself, shipping costs, and disposal costs. The greatest contribution comes from the cost of the material itself, which is $130/L. Shipping costs will vary widely based on where the printer is being used and where the photopolymer resin is being produced. In this analysis we determined a shipping rate of $4.42/L, assuming the resin was being produced in the eastern United States [33] and shipped in 20L barrels to the sponsor company in Vancouver, Canada. The support material for a printed part is waste material since it is a thermoset plastic and cannot be recycled, therefore disposal costs must also be considered. The waste removal costs for a small company in North Vancouver was estimated to be $0.37/kg. See *Appendix C – Economic Analysis Details* for more information on these calculations.

Total material costs were calculated by:

$$\begin{aligned} &total\ material\ cost \\ &\qquad = (part\ volume + support\ volume) * (material\ cost + shipping\ cost \\ &\qquad + garbage\ disposal\ cost) \end{aligned}$$

The power costs include the power consumed by the projector, the fans in the printer, and the UV post-cure chamber. The power input for the projector is 200 W and it typically runs at three-quarters power for 50% of the total print time [34]. The power input for the printer fans and the UV post-cure chamber are 18 W and 44 W, respectively [35, 34]. The typical power rates in Vancouver [36] were used to estimate the total power costs for a part. This cost was found to be negligible compared to the labour and material costs for both the small-scale and large-scale operation.

Total power costs were calculated by:

$$\begin{aligned} &total\ power\ cost \\ &\quad = [(projector\ power * 50\%\ on\ time * 0.75\ capacity + fan\ power) * print\ time \\ &\quad + (post\ cure\ power * post\ cure\ time)] * power\ cost \end{aligned}$$

The difference between the small-scale and large-scale operation impacts the labour costs of each part. The small-scale operation has one highly-skilled operator, supervising one print at a time. The large-scale operation can run multiple prints at a time in an assembly-line fashion.

Labour costs for the small-scale operation were calculated using an estimated wage of the operator of $80/hr, as recommended by our sponsor company [33].

Total labour costs were calculated by:

$$\begin{aligned} &small\ scale\ labour\ cost \\ &\quad = (print\ time + prep\ and\ clean\ up\ time + IPA\ wash\ time + UV\ post\ cure\ time \\ &\quad + support\ removal\ time) * operator\ wage \end{aligned}$$

To calculate labour costs for the large-scale operation, first the number of operators needed to maintain continuous production was estimated. It was assumed that the large-scale operation makes use of multiple printers at the same time and is able to post-cure multiple parts at the same time as well. In the time it takes to print the case study part used for the economic analysis, approximately 4 operators are needed per part to complete the other steps of prep/clean-up, IPA wash, and support removal. These production operators are paid $15/hr based on Pay Scale Canada [37].

$$large\ scale\ labour\ cost = print\ time * number\ of\ operators\ needed * operator\ wage$$

The other benefit of reducing supports is the reduction in the time it takes for an operator to remove the supports once the part is printed. This process varies greatly from part to part. With simple

geometries, it may be as simple as snapping off all the supports at once. However, for complex geometries or for parts that require a high-quality surface finish, it could involve carefully snipping off each support individually. In our analysis, we estimated the time to remove the supports from the original support model to be 5 minutes, and from the reduced support model to be 1 minute.

Since 3D printed parts vary drastically in shape, volume, and print time, it is difficult to decide on an "average part". Numbers in this economic analysis are based on a case study of a part designed over the course of this project, discussed more in *Section 9.3*. The cost break-down for this part for each model is shown in

Figure 22. Both models show labour contributing to the majority of the cost. In the small-scale model, only 3% of the cost comes from the material, while in the large-scale model this is increased to 22%. This shows that reducing supports has a greater potential to reduce overall costs for the large-scale operation than for the small-scale operation.

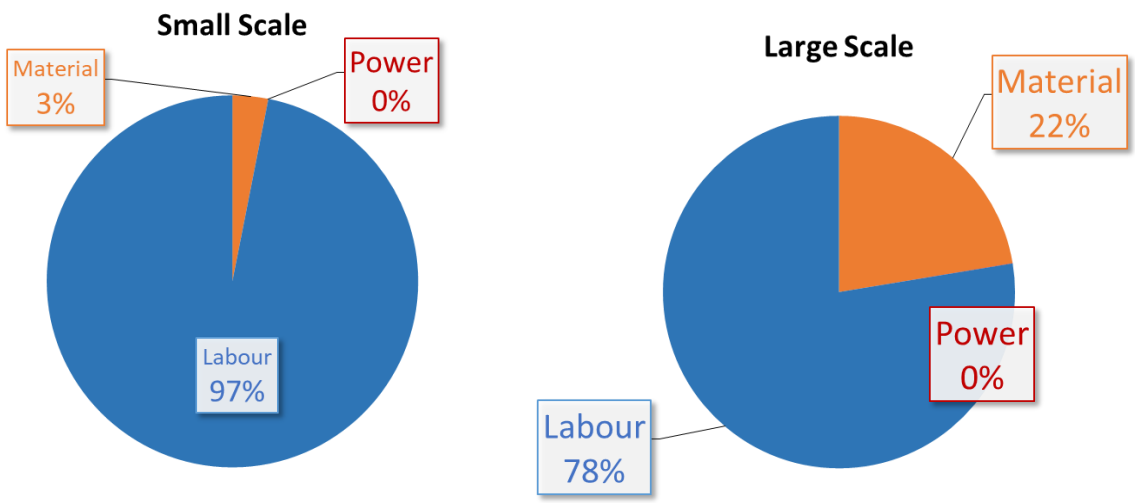

Figure 22: Cost break-down for small scale operation (left) vs. large scale operation (right)

9.2 Environmental Analysis

The environmental analysis was split into three categories: material production, transportation, and 3D printing, with the CO_2 and energy footprint determined for each.

Material production data was collected from *Plastics Europe* [38], a European plastics industry association who performs lifecycle analyses of plastics components. We found that photopolymer resins consist of 50-80 wt% binders, 10-40 wt% monomers and a small amount of photo-initiators [39]. Data

corresponding to two common binders, methyl methacrylate (MMA) and styrene, as well as two common monomers, glacial acrylic acid (GAA) and methacrylic acid (MA), were averaged and used in the analysis.

We made several assumptions regarding the material production. We assumed the amount of photo-initiators to be negligible and excluded them from our analysis. For the binder and monomer composition, we assumed them to be 70 wt% and 30 wt%, respectively. We also neglected the environmental impact of converting the components into a photopolymer resin, since this is only one step in a long materials manufacturing process and is essentially a mixing step.

The environmental impact of the material production of the photopolymer resin was assessed based on several categories, including energy use, air emissions, water consumption, and solid waste. The breakdown of each of these categories into their subcomponents is provided in Appendix D – Environmental Analysis Details.

CO_2 equivalent emissions were calculated using the global warming potentials in Table 6 below, which standardize various emissions against CO_2 emissions. These potentials are obtained from the *Greenhouse Gas Protocol,* an organization that maintains the most widely used standards related to greenhouse gas emissions [40].

Table 6: Global warming potentials

Emission	Global Warming Potential
CO_2 (fossil & biogenic)	1
CO	2
CH_4	25
SO_2	-
NO_x	298

Transportation data was generated using CES Eco-Audit software. Our sponsor orders their resin from a supplier located on the east coast of the United States, therefore we assumed that resin would be delivered by truck over a distance of 5000 km. Our results are summarized in *Appendix D – Environmental Analysis Details*.

To understand the environmental impacts of SLA-DLP printing we estimated power consumption from our sponsor and converted the energy use to CO_2 emissions using data from BC Hydro.

Combining these numbers, we found that more than 99% of the environmental impact of SLA-DLP printing comes from the material production of the photopolymer resin. Therefore, any reduction of material through support reduction will directly translate to the reduction of the environmental impact of the SLA-DLP printing process.

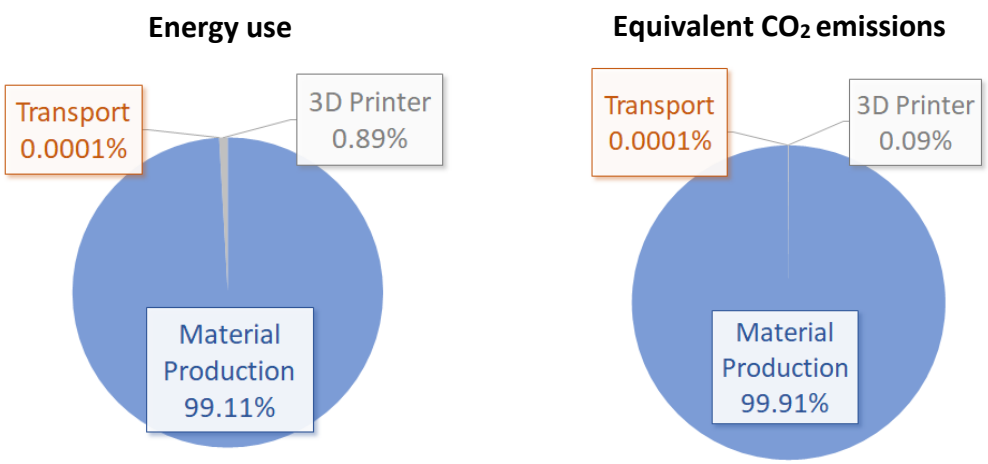

Figure 23: Environmental impact of SLA-DLP printing.

9.3 Socio-Economic Case Study

After compiling the economic and environmental factors which affect SLA-DLP printing we performed a case study on our simplest test piece in order to estimate the potential impact of reducing supports. Using the calculations described in *Section 8* of this report, we found that brittle failure was the limiting factor for this test piece and required supports to be placed every 11 mm. Using our sponsor's support generation software, we compared the resulting support density to our sponsor's default support spacing of 2 mm.

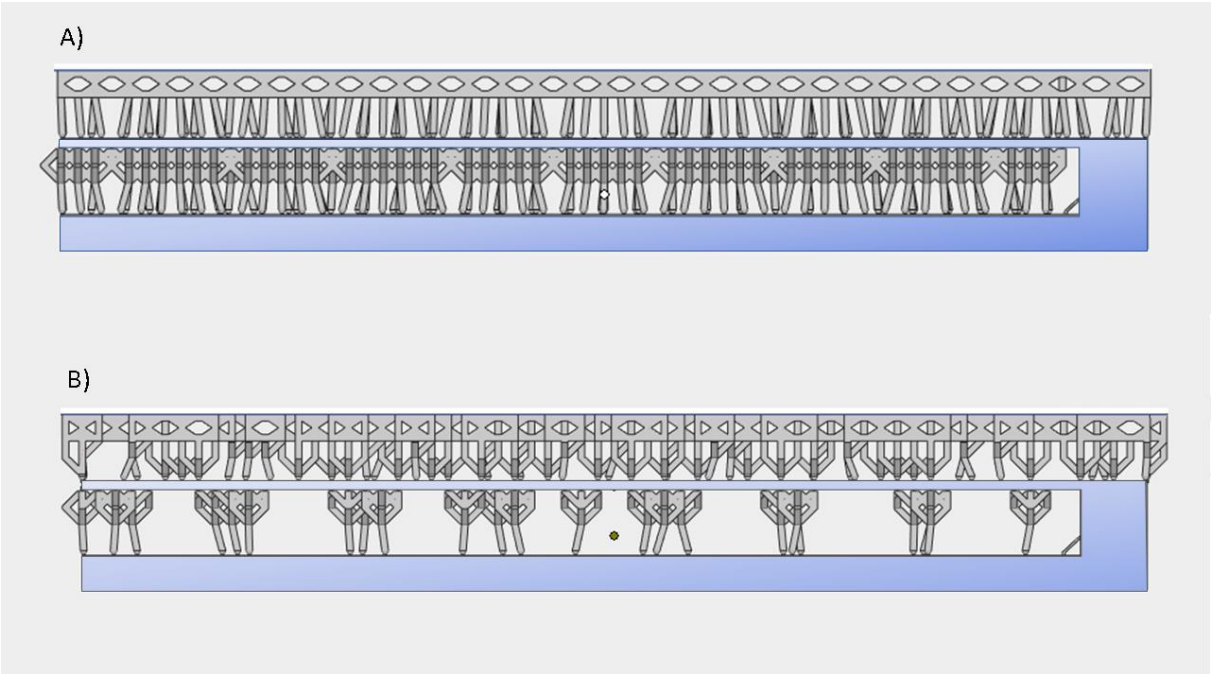

Figure 24: Case study part with supports generated by the original parameters (A) and by the models developed from this project (B)

The reduction in supports for our case study part resulted in a 16% reduction in material. This resulted in a 16% reduction of the environmental impact as well as an 8% reduction in overall costs for a large-scale operation or a 4% reduction in overall costs for a small-scale operation. These numbers are dependent on the ratio of the volume of supports to the volume of the part. For parts with many supports this reduction will be larger and for parts which are largely unsupported, this value will be smaller.

10Summary

In summary, we have developed three models pertaining to the mechanical failure of SLA-DLP parts during printing:

1. An elastic deflection model;
2. A right-angle brittle failure model; and,
3. A variable angle brittle failure model.

Together these models have shown that there are few cases where the current practice of placing supports every 2 mm is necessary within the scope of this capstone. Since it is not necessary for supports to be placed as densely as current practice dictates, there is potential for support reduction. Our socio-economic assessment has shown that reducing the number of supports results in significant monetary savings and reduces the negative environmental impact of SLA-DLP printing. Overall, this project provides evidence that more research should be done into reducing supports in SLA-DLP printing. The next steps that we believe should be taken are outlined in our recommendations.

11 Recommendations

Through the literature review, development of our support generation model, and conversation with our sponsor company and project supervisor, Professor Sinclair, we have concluded that most printed parts are over-supported based on the current practices for support generation. The two major forces acting on the part during printing, self-weight and separation force, are dependent on the part geometry and material. Therefore, using set parameters to generate supports for all geometries and materials results in more supports than necessary.

To improve support placement in SLA-DLP printing, we recommend that the following take place:

1. The test pieces we developed should be printed and compared to the expectations based on our assumptions and calculations.
2. Deviations from our models which arise from the verification prints should be addressed and the models adjusted as necessary.
3. The scope of the project should be expanded to encompass all possible failure modes rather than just the mechanical failure of the print itself (e.g., separation from the supports).
4. Amount of supports should be reduced based on our calculations in order to reduce the cost of printing as well as reduce the environmental impact of printing.
5. Support generation software should be improved to account for part geometry and material properties of the part being printed.

12 Project Work Summary

The purpose of this work summary is to describe the project planning process, task breakdown, and specifically who was responsible for each task. This section will also provide details of challenges that the group had while completing the project. During project planning, week-by-week milestones were laid out during the proposal report and midterm report. Our group closely followed these milestones to ensure that all worked was completed on time.

Before the midterm, our group worked together to define our project and set our project objective and goals. This started by completing an in-depth technical review to understand what research had been previously done and what calculations existed for us to build our project off. All group members researched literature and compiled sources into one document. For each source, group members wrote a brief summary of the articles and how it could add to our project. After reviewing all the researched literature, we define the scope of our project. This included the project objectives, constraints, and free variables. After defining the problem, we were able to move on to the project objectives and quantitative goals. Our main goal was to use mechanics to derive equations based on the stresses and failure modes defined from our literature review. Using our derived equations including valid assumptions, our goal was to design two test pieces that could verify our calculations to be used by the sponsor company.

To achieve the goals by the midterm, tasks were done concurrently by two groups of three team members. James, Kaiya, and Emma were responsible for starting the calculations for the self-loading stress and failure by excessive deflection. This involved researching literature and using solid mechanics equations to derive equations based on assumptions. With that equation, the overhang test piece was designed. In addition, an FDM (Fused Deposition Model) test piece was designed to verify the process of measuring the deflection of the test piece. Elizabeth, Max, and Eva were responsible for beginning the socio-economic study. They started with a comprehensive literature review to break up the economic and environmental factors that relate to support reduction. In addition, Max and James began calculations based on mechanics for stress caused by separation force.

After the midterm, tasks were done concurrently by groups of team members to achieve our goals by the final. James and Max finished the calculations for stress caused by separation force. The group worked together to create the brittle fracture test piece based on the calculations. James modeled the test piece using SolidWorks. Kaiya was responsible for communication between our team and our

sponsor company. He sent both the brittle fracture test piece and the self-load test piece to our sponsor company to be printed. Kaiya, Emma, and James worked together to apply our previous calculations from brittle fracture and self-load to derive stress calculations for critical angle to see the effect of printing at an angle. Max helped to create a process map for the critical angle calculations. Eva, Max, and Elizabeth worked together to complete the economic and environmental analysis. Eva focused on the economic analysis, while Elizabeth and Max focused on the environmental analysis. The groups of team members ensured that their scope of work is completed, but the team as a whole worked together to support all the project goals.

Throughout the project, our group encountered multiple challenges that our group had to solve together. The biggest challenges that occurred was that we couldn't get our test pieces printed before the final. Although we had our test pieces ready shortly after the midterm, the test pieces were unable to be printed. After sending the sponsor company our original test pieces, we learned that our test pieces were too large for the print bed. We then had to redesign the test pieces and resend them to the sponsor company. Near the end of the project, our representative from the sponsor company didn't have time to print all the group's test pieces. In order to overcome this obstacle, we need to assume that the assumptions that we made for our calculations were correct. In future work, we would like to have the test pieces printed to be able to verify our calculations and recommendations. Another challenge throughout the project was making appropriate assumptions to help us limit our project scope. The assumptions were necessary for both calculations and designing test pieces, but also the socio-economic assessment. We couldn't have access to the SLA-DLP printer, which required us to make many assumptions about the printing process. Our group worked hard to research literature and consult our project supervisor and sponsor company to come up with appropriate assumptions to help us reach our project goal.

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14 Appendix A – Mechanical Properties

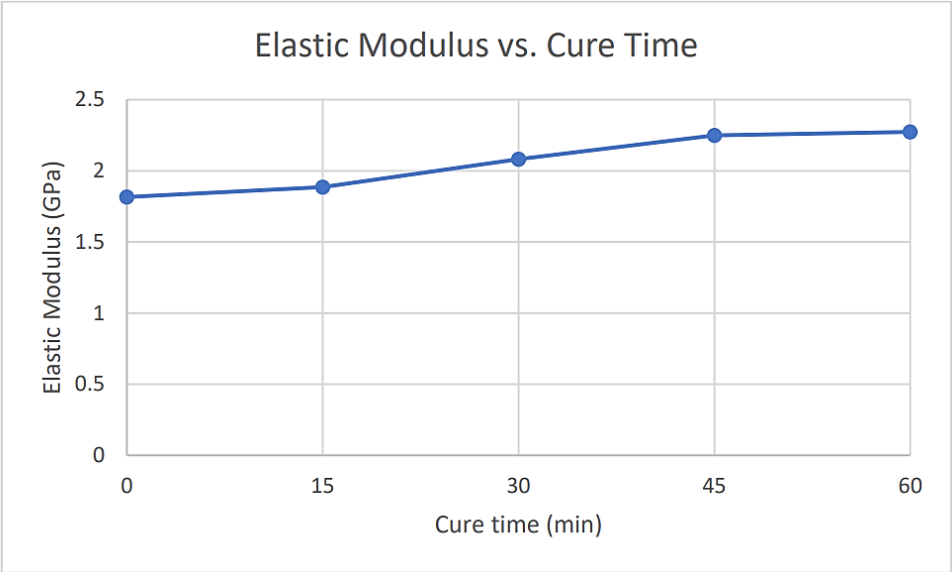

Figure 25: Elastic modulus vs. cure time of P-121 Resin [28].

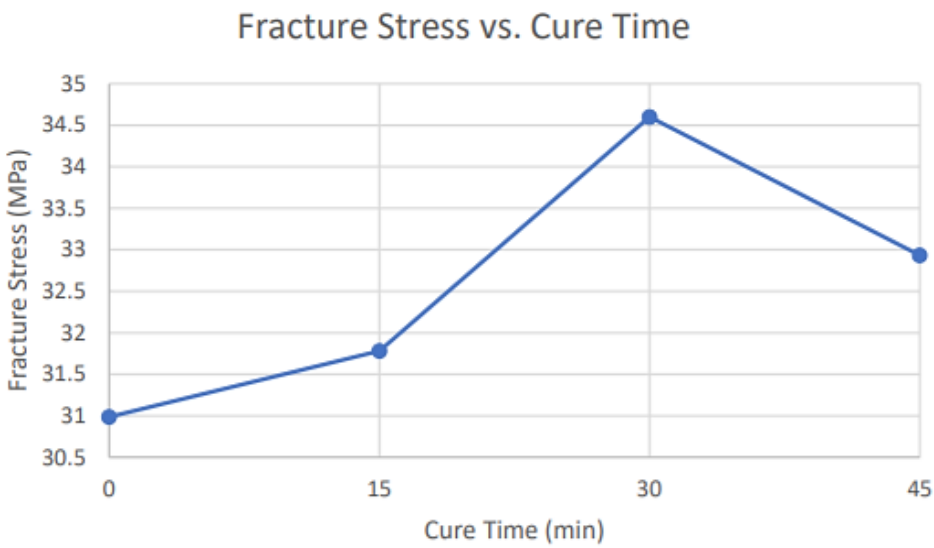

Figure 26: Fracture stress vs cure time of P-121 resin [28].

15 Appendix B – Stefan Adhesion

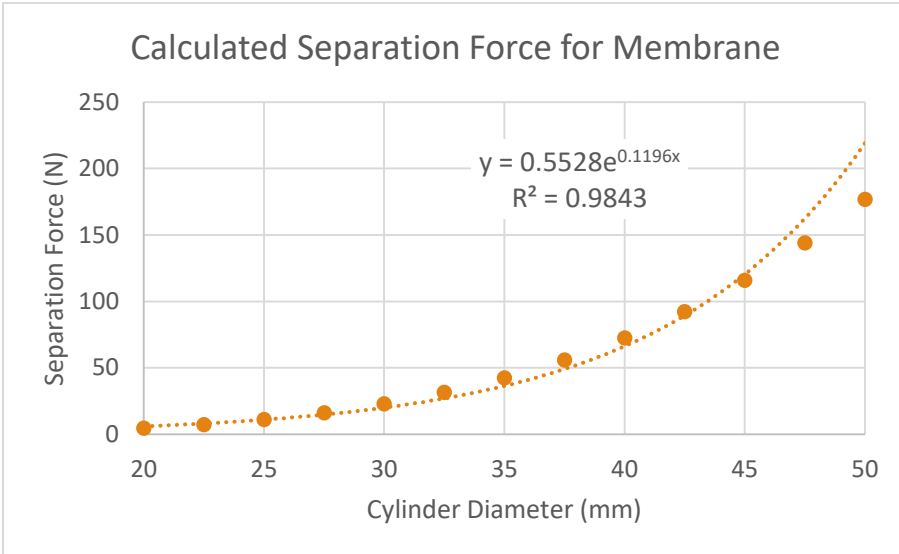

Figure 27: Our calculated values for Stefan adhesion.

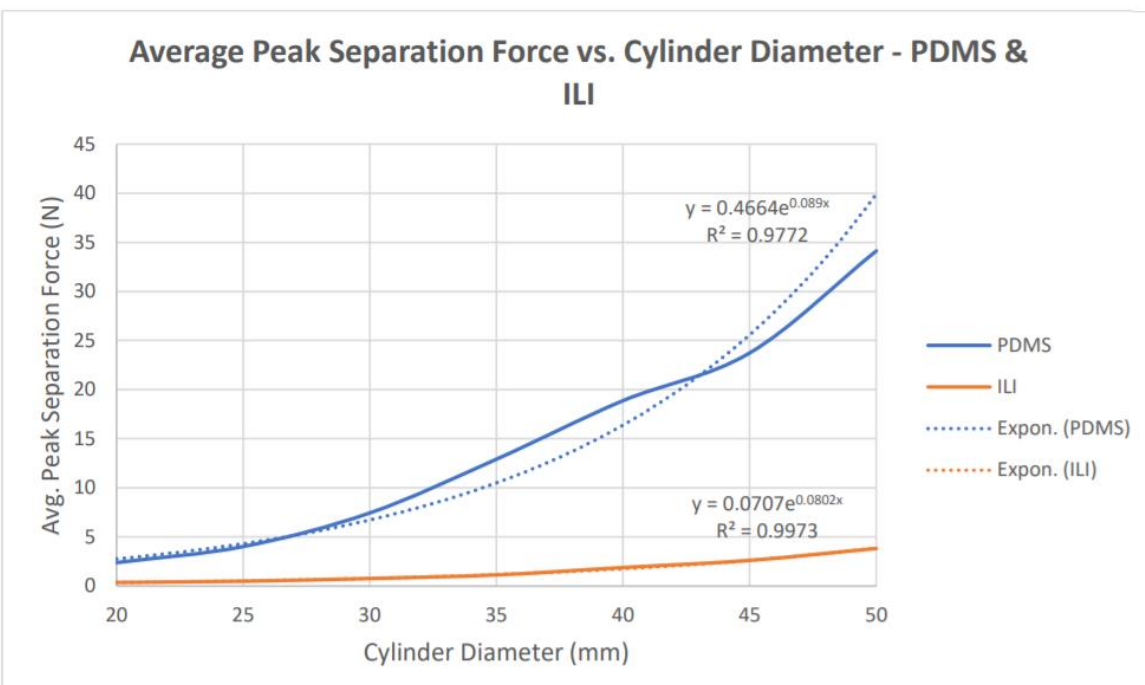

Figure 28: NewPro3D's empirical data for separation force, showing the relationship between the average peak separation force and the cylinder diameter [23].

16 Appendix C – Economic Analysis Details

This appendix will show the methods used for determining various values used in the economic analysis.

Given from Sponsor Company
Calculated/retrieved

Category	Description	Value	Unit Rate
Time	IPA wash	0.25	hr
	UV post-cure	0.5	hr
	Prep/clean-up	0.75	hr
Material	P-121	130	$/L
	Shipping	$4.42	$/L
	Garbage disposal	$0.37	$/kg
Power	Projector	200	W
	Fans	18	W
	Post-Cure Power	44	kWh
	Power cost	0.1326	$/kWh
People	Operator (small-scale)	80	$/hr
	Operator (large-scale)	15	$/hr

Figure 29: Values used for economic analysis

Shipping rates were calculated based on a cost estimation feature provided by FedEx. The flowing assumptions were made for the cost estimation:

- Shipping from the eastern USA – zip code: 07201
- Shipping to Vancouver, Canada – postal code: V7L 1B3
- Shipping in a 20L barrel – height: 37 cm, diameter: 30 cm
- Neglect customs costs

The cost of shipping one barrel of resin is $88.44, which equates to 4.42/L [41].

Garbage disposal costs were estimated by using a cost estimation feature provided by Waste Management Canada. The following assumptions were made:

- 2-yard trash dumpster required - Capacity: 400 lbs
- Pick-up once per week
- 75% filled each week

The monthly cost for garbage disposal is $204.21, which equates to $0.37/kg [42].

17 Appendix D – Environmental Analysis Details

Table 7: Values used in environmental analysis

	Binders			Monomers			Photopolymer
	Styrene	**MMA**	**Average**	**Glacial Acrylic Acid (GAA)**	**Methacrylic Acid (MA)**	**Average**	**Weighted Average**
Primary Energy Demand [MJ/kg]							
Energy consumption; renewable & non-renewable	82.6	65.5	74.05	45.23	59.23	52.23	67.504
Air Emissions [kg/kg]							
CO_2 (fossil & biogenic)	2.4	3.21	2.805	1.1028	1.5651	1.33395	2.363685
CO	0.0036	8.02E-04	0.002201	0.0022	0.0142	0.0082	0.0040007
CH_4	0.03	0	0.015	0.00412	0.00435	0.004235	0.0117705
SO_2	0.0065	1.37E-02	0.0101	0.00151	0.00241	0.00196	0.007658
NO_x	0.0043	5.05E-03	0.004675	0.00102	0.0021	0.00156	0.0037405
Total CO2 equivalent	4.44	4.72	4.58	1.51	2.33	1.92	3.78
Water Consumption [kg/kg]							
Cradle-to-gate fresh- and seawater	143.00	500.00	321.50	498.18	902.75	700.47	435.1895
Solid Waste [kg/kg]							
Incineration	0.02000	0.00000	0.01000	0.01690	0.03110	0.02400	0.0142
Landfill	0.00000	0.15600	0.07800	0.00091	0.00000	0.00045	0.05473635
Recovery	0.00000	0.00000	0.00000	0.00013	0.00000	0.00006	0.00001875
Total	0.02000	0.15600	0.08800	0.01793	0.03110	0.02452	0.0689551

Table 8: CO_2 emissions calculations

Energy	4.1	MJ
CO_2	0.295	kg/Trip
Container volume	105	m^3
CO_2	0.00000255	kg CO_2/Kg resin
Energy	0.0000355	MJ/kg

In the table above CO_2 emissions per kg of resin are calculated by first dividing the CO_2 per trip by the container volume, resulting in the CO_2 per m^3 then using the density of the resin, converting this into CO_2 per kg of resin. Energy per kg of resin is calculated in a similar fashion.

Table 9: Power Consumption Environmental Impact

Parameter	Value	Unit
Volume of resin	0.0005	m^3
Mass of resin	0.56	kg
Printing energy	168	kWh
Post cure energy	0.088	kWh
Total energy/Kg resin	0.6054	MJ/Kg
Carbon emissions/energy use	0.0056	kg CO_2/MJ
Total carbon emissions	0.0034	kg CO_2/kg resin

The volume of resin in the table above is the volume of the test piece used in the case study discussed in the Socio-Economic Case Study, this was converted to a mass using the density of the resin. The printing time was calculated using the height of the part and the sponsor's printing speed. The energy was then calculated using the power consumption of the sponsor's printer and the printing time. Post cure energy was calculated based off of the power consumption of the post cure chamber and the post cure time.

18 Appendix E – Deflection Calculations

18.1 Overhang Deflections

Deflections of the overhangs of the deflection test piece were performed using equation (14). The following is a sample calculation for deflection with length of 10 cm and thickness of 1 mm:

$$\delta_{max} = \frac{3\rho g c^4}{2Ea^2} \tag{13}$$

$$\delta_{max} = \frac{3(1080kg/m^3)(9.81m/s^2)(0.1m)^4}{2(1.8 * 10^9 Pa)(0.001m)^2}$$

$$\delta_{max} = 0.0008829m \; or \; 0.8829mm$$

18.2 Maximum Stress Acting on the Deflection Test Piece During Printing

The deflection test piece has a large surface area therefore it was necessary to check that the separation forces acting on it would not result in stresses exceeding the fracture stress of the material.

The average peak separation force, from Figure 28 is as follows:

$$F = 0.0707 \exp(0.0802 * D) \tag{14}$$

Since separation forces are reported as a function of cylinder diameter, part areas must be converted to equivalent cylinder diameters.

$$D = 2(\frac{A}{\pi})^{1/2} \tag{15}$$

Using the cross sectional area of the part (4970 mm^2) and inserting (15) into (14) we find the following:

$$\boldsymbol{F = 0.0707\, exp\left(2 * 0.802\left(\frac{4970}{\pi}\right)^{\frac{1}{2}}\right) = 24,600N} \tag{16}$$

This results in a stress of:

$$\sigma = \frac{F}{A} = \frac{24594\, N}{(4970 * 10^{-6})m^2} = 4.9\, MPa$$

Since 4.9 MPa is much less than the green fracture stress of 31 MPa, which we found from Figure 26, the part will not fracture during printing.

19 Appendix F – Critical Angle Calculations

Like the case for brittle fracture, there are two forces acting on the beam: the weight of the beam and the separation force being applied across the horizontal width. The separation force of brittle fracture F_b has already been found in equation (9).

$$F_b = 0.0707\exp(1.604 * \sqrt{ab})$$

In the angled case, the area in contact with the membrane is no longer ab. It is now it is wb.

$$\mathrm{w} = \frac{a}{\sin(\theta)} \tag{17}$$

Combining (**17**) with (9) we find the magnitude of the separation force as

$$F_s = 0.0707\exp(1.604 * \sqrt{wb})$$

$$F_s = 0.0707\exp(1.604 * \sqrt{\frac{ab}{\sin(\theta)}}) \tag{18}$$

The position of F_s is placed in the center of w at the position P_s,

$$P = X - \frac{w}{2}$$

X can we written in common terms,

$$X = c\cos(\theta) \tag{19}$$

$$P_s = c\cos(\theta) - \frac{a}{2\sin(\theta)} \tag{20}$$

Since the self-weight of the beam is treated as constant throughout the part, we can convert it to a point load in the center of its length.

$$P_g = \frac{X}{2}$$

$$P_g = \frac{c\cos(\theta)}{2} \tag{21}$$

Knowing the magnitudes and positions of the forces, we can determine the maximum moment in the part. This magnitude occurs at the length of the beam, at position X.

$$M = M_s + M_g$$

Combining (8), (**18**), (20) and (**21**) while knowing that the bending force in the component acting perpendicular to the beam, we find

$$M = (F_s * \cos(\theta)\, P_s + F_g * \cos(\theta)\, P_g)$$

$$M = \frac{\rho g a b c^2 \cos^2(\theta)}{2} + \left(c\cos(\theta) - \frac{a}{2\sin(\theta)}\right)\left(\cos(\theta) 0.0707 \exp\left(1.604\sqrt{\frac{ab}{sin(\theta)}}\right)\right) \tag{22}$$

The table below summarizes the magnitude and position of the forces.

Table 10: Magnitude and position of the self weight and separation forces for the angled support model

Force	Magnitude	Position as a Point load
Self-Weight	$F_g \cong \rho g a b c$	$P_g = \frac{c\cos(\theta)}{2}$
Separation	$F_s = 0.0707\exp(1.604 * \sqrt{\frac{ab}{\pi\sin(\theta)}})$	$P_s = c\cos(\theta) - \frac{a}{2\sin(\theta)}$

The maximum stress in the part is a function of the total moment M as well as the section modulus Z. According to "Materials Selection in Mechanical Design" Appendix 4 [21], we have the following relations,

$$\sigma_{bend} = \frac{M}{Z} \tag{23}$$

$$Z = \frac{bh^2}{6} \tag{24}$$

From Figure 20, we find h in terms of θ

$$h = \frac{a}{\cos(\theta)} \tag{25}$$

Combining (**22**), (*23*), (**24**) and (**25**) we find the bending stress in the part, in terms of the support

$$\sigma_{bending} = \frac{6\cos^2(\theta)}{ba^2}\left[\frac{\rho gabc^2\cos^2(\theta)}{2} + \left(c\cos(\theta) - \frac{a}{2\sin(\theta)}\right)\left(\cos(\theta) * 0.0707\exp\left(1.604\sqrt{\frac{ab}{sin(\theta)}}\right)\right)\right] \tag{26}$$

Now that the bending stress is written in terms of the critical angle, the tensile stress must also be found. The forces being applied are the contributions running parallel with the beam.

$$\sigma_{tensile} = \frac{F_{parallel}}{A}$$

$$\sigma_{tensile} = \frac{\sin(\theta)\left(F_g + F_s\right)}{ab} \tag{27}$$

Inserting (8) and (**18**), the magnitudes of F_g and F_s into (27)

$$\sigma_{tensile} = \frac{\sin(\theta)\left(\rho gabc + 0.0707\exp(1.604 * \sqrt{\frac{ab}{\pi\sin(\theta)}})\right)}{ab} \tag{28}$$

So the top of the beam in Figure 20 is experiencing tension along its longitudinal axis from two stress components, the bending stress and the tensile stress.

$$\sigma_y = \sigma_{tensile} + \sigma_{bending}$$

Inserting (**26**) and (**28**) into the above sum of stresses, we have the yield stress acting parallel to the longitudinal axis of the beam, at the upper surface of the part.